

SEQUENTIAL GAUSSIAN APPROXIMATION FOR EMPIRICAL DISTRIBUTION PROCESSES OF DEPENDENT CAUSAL TIME SERIES

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We develop a strong sequential Gaussian approximation theory for empirical distribution processes of stationary causal time series. The process is indexed jointly by time and fixed-dimensional lower orthants, so the approximation describes the evolving empirical law rather than only the terminal empirical distribution. A central feature is that the assumptions are imposed through physical dependence of the causal data-generating recursion, rather than through classical mixing conditions such as absolute regularity, strong mixing, or S-mixing. This provides a verification route for nonlinear recursions and scale processes where mixing conditions can be difficult to check. Under geometric physical dependence and local boundary regularity, we construct a covariance-matched Gaussian process whose uniform coupling error is polynomially smaller than the natural square-root sample-size scale. We also give a polynomial-dependence result under an explicit decay condition on the physical-dependence exponent. A sharp local oscillation bound for dependent lower-orthant empirical processes is developed uniformly over all threshold locations. Its leading fluctuation term matches the classical independent order, while an additional predictable boundary term isolates the effect of serial dependence. The approximation yields Gaussian bridge, change-point, endpoint distributional, scalar quantile, and dependent multiplier calibration results, supported by simulations and a financial-stress application.

1. Introduction. This paper studies a strong Gaussian approximation problem for the sequential empirical distribution process of a dependent time series. Let $(X_i)_{i \in \mathbb{Z}}$ be a stationary causal process in $\mathbb{R}^d$, with d fixed, and write inequalities coordinatewise. The process of interest is

$$(1.1) \quad S_n(u, s) = \sum_{i=1}^{\lfloor nu \rfloor} \{\mathbf{1}(X_i \leq s) - F(s)\}, \quad (u, s) \in [0, 1] \times \mathbb{R}^d,$$

where $F(s) = \mathbb{P}(X_0 \leq s)$. We construct, on an enlarged probability space, a centered Gaussian process T_n indexed by $[0, 1] \times \mathbb{R}^d$ with covariance

$$(1.2) \quad \text{Cov}\{T_n(u, s), T_n(v, t)\} = n(u \wedge v)\Gamma(s, t), \quad \Gamma(s, t) = \sum_{h \in \mathbb{Z}} \text{Cov}\{f_s(X_0), f_t(X_h)\},$$

where $f_s(x) = \mathbf{1}(x \leq s) - F(s)$. The aim is the uniform coupling

$$(1.3) \quad \sup_{u \in [0, 1]} \sup_{s \in \mathbb{R}^d} |S_n(u, s) - T_n(u, s)| = o_{\mathbb{P}}(\sqrt{n}).$$

Thus the approximation is pathwise, sequential in u , and uniform over all lower-orthant thresholds s . The covariance in (1.2) is the long-run covariance of the dependent indicator process, and the factor $u \wedge v$ is the sequential clock.

MSC2020 subject classifications. Primary 62M10, 60F17; secondary 62G30, 62G10, 62G20.

Keywords and phrases. Sequential empirical process, Gaussian approximation, physical dependence, lower orthants, change-point analysis, dependent multiplier bootstrap.

This object appears in several forms of distributional inference. Bridge statistics for distributional change and monitoring are obtained by subtracting a terminal or historical estimate from $S_n(u, \cdot)$ (Csorgo and Horvath, 1997; Inoue, 2001; Aue and Kirch, 2024; Kojadinovic and Verdier, 2021; Holmes, Kojadinovic and Verhoijesen, 2024). Recent online procedures also emphasize pathwise calibration and post-detection uncertainty (Chen, Wang and Samworth, 2024). In dimension one, the inverse map connects the same sequential process to recursive quantiles and Bahadur–Kiefer type expansions (Bahadur, 1966; Kiefer, 1967; Wu, 2005a). In fixed dimension, lower-orthant and rectangle probabilities enter joint-tail inference, empirical copula methods, and copula-based time-series diagnostics (Berghaus and Bücher, 2017; Lu and Zheng, 2020; Nagler, Krüger and Min, 2022; Kojadinovic and Yi, 2024). Dependent bootstrap methods for related empirical-process and goodness-of-fit statistics include Buehlmann (1994), Buecher and Kojadinovic (2016), Doukhan et al. (2015), and Chandy et al. (2026). In each case, a Gaussian approximation with covariance Γ and error $o_{\mathbb{P}}(\sqrt{n})$ gives a direct route from a dependent empirical statistic to its Gaussian analog.

The independent empirical-process theory provides the classical benchmark. For iid observations, after the probability integral transform $U_i = F(X_i)$, and with F_n denoting the empirical distribution of $U_1, \dots, U_n$, the Komlos–Major–Tusnady construction gives a Brownian bridge B_n such that

$$\sup_{t \in [0,1]} |\sqrt{n}\{F_n(t) - t\} - B_n(t)| = O_{\text{a.s.}}\{n^{-1/2} \log n\}.$$

Equivalently, on the unnormalised scale used in (1.1),

$$\sup_t |S_n(1, t) - \sqrt{n}B_n(t)| = O_{\text{a.s.}}(\log n).$$

This logarithmic strong error is the one-dimensional iid reference point for Brownian bridges, Kiefer processes, and Bahadur–Kiefer expansions (Kiefer, 1972; Komlós, Major and Tusnady, 1975, 1976; Shorack and Wellner, 1986; Csorgo et al., 1986; van der Vaart and Wellner, 1996). Modern Gaussian approximation theory replaces exact pathwise KMT embeddings by nonasymptotic approximation of distributional functionals. For example, for independent M -dimensional vectors Y_i and a matching Gaussian vector Z , a representative high-dimensional bound in Chernozhukov, Chetverikov and Kato (2013) has the form

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left(\max_{1 \leq j \leq M} n^{-1/2} \sum_{i=1}^n Y_{ij} \leq x \right) - \mathbb{P} \left(\max_{1 \leq j \leq M} Z_j \leq x \right) \right| \lesssim \left\{ \frac{\mathfrak{B}_n^2 \log^7(Mn)}{n} \right\}^{1/6},$$

under suitable moment or envelope conditions, where $\mathfrak{B}_n$ denotes the corresponding envelope scale. The empirical-process results of Chernozhukov, Chetverikov and Kato (2014) further approximate suprema $\sup_{f \in \mathcal{F}_n} \mathbb{G}_n f$ by Gaussian suprema and, in local empirical applications, give explicit rates in powers of the local effective sample size. These results provide the metric and nonasymptotic reference point for the present paper, although our object is a pathwise sequential lower-orthant empirical process generated by a dependent causal sequence.

For time series, the rate also depends on how serial dependence is represented and on which covariance is matched. The physical-dependence framework of Wu (2005b) expresses dependence through the causal data-generating recursion and has been used for martingale approximations, spectral inference, robust M -estimation, empirical distribution functions, and strong invariance principles for nonlinear time series (Wu and Woodroffe, 2004; Shao and Wu, 2007; Wu, 2007a, 2008, 2007b; Berkes, Liu and Wu, 2014). For vector-valued time-series sums, Zhang and Wu (2017) establish Gaussian approximation for

$$\left| n^{-1/2} \sum_{i=1}^n X_i \right|_{\infty}$$

with rates expressed through dependence-adjusted norms, moment exponents, and $\log M$, where M is the vector dimension. Sequential high-dimensional versions are developed by [Mies and Steland \(2023\)](#). For univariate nonstationary sums with $p > 2$ moments, KMT-type approximations by Gaussian analogues Y_i can reach the sharp order

$$\max_{1 \leq k \leq n} \left| \sum_{i=1}^k X_i - \sum_{i=1}^k Y_i \right| = O_{\mathbb{P}}(n^{1/p}) \quad \text{or} \quad O_{\mathbb{P}}(n^{1/p} \log n),$$

depending on the covariance-matching and independent-increment construction; see [Bonnerjee, Karmakar and Wu \(2024\)](#). Martingale coupling methods provide another closely related route, for example [Cattaneo, Masini and Underwood \(2025\)](#). These works concern sums, maxima, martingales, or partial-sum processes. The present paper develops the corresponding lower-orthant empirical-distribution approximation, where the coordinates are the discontinuous indicators $x \mapsto \mathbf{1}(x \leq s)$ and the approximation must be uniform in both sequential time and threshold.

The dependent empirical-process literature supplies the second reference point. Weak convergence and invariance principles under mixing, weak dependence, and long-memory conditions include [Billingsley \(1968\)](#), [Withers \(1975\)](#), [Mehra and Rao \(1975\)](#), [Doukhan, Masart and Rio \(1995\)](#), [Shao and Yu \(1996\)](#), [Rio \(2000\)](#), [Dehling and Taqqu \(1989\)](#), and [Ho and Hsing \(1996\)](#). Residual and time-series empirical processes have been analyzed, for example, by [Chan and Ling \(2008\)](#). Sequential empirical processes and sequential empirical copula processes under serial dependence were developed by [Inoue \(2001\)](#), [Buecher \(2015\)](#), [Buecher and Volgushev \(2013\)](#), and [Buecher and Kojadinovic \(2016\)](#). These results identify the correct Gaussian limits for many dependent empirical statistics. The stronger question addressed here is whether one can construct, with an explicit rate, a Gaussian process on the same probability space that is uniformly close to the whole sequential empirical distribution process.

Known strong approximations show why this question is delicate. Classical mixing approaches, including [Philipp and Stout \(1975\)](#), [Berkes and Philipp \(1979\)](#), and [Dudley and Philipp \(1983\)](#), establish Kiefer-process approximations under specific structural assumptions on the mixing coefficients. In the weak-dependence direction, [Berkes, Hörmann and Schauer \(2009\)](#) work under S -mixing conditions, which are well suited to functionals of iid sequences and require approximation by finite-dependent variables together with deviation control for the approximation error. Writing

$$R_X(s, t) = \sum_{1 \leq i \leq t} \{\mathbf{1}(X_i \leq s) - F(s)\}$$

and denoting by K_X the covariance-matched Kiefer process, their strong approximation has the benchmark form

$$\sup_{s \in \mathbb{R}, t \in [0, 1]} |R_X(s, \lfloor nt \rfloor) - K_X(s, \lfloor nt \rfloor)| = O_{\text{a.s.}}\{\sqrt{n}(\log n)^{-\lambda}\}$$

for some $\lambda > 0$. Thus the error is below the empirical-process scale, but by a polylogarithmic factor. Such a bound is sufficient for functional limit and iterated-logarithm consequences, while a polynomial coupling rate is needed to track the rate contribution of truncation, blocking, off-grid oscillation, and covariance matching.

The strongest polynomial-rate results for dependent empirical distribution functions use another dependence language. In the scalar case, [Dedecker, Merlevede and Rio \(2013\)](#) prove that, if the indicator-level coefficient satisfies

$$\beta_{2,X}(k) = O(k^{-1-\delta}) \quad (\delta > 0),$$

then a Kiefer process K_X can be constructed so that

$$\sup_{s \in \mathbb{R}, t \in [0,1]} |R_X(s, \lfloor nt \rfloor) - K_X(s, \lfloor nt \rfloor)| = O_{\text{a.s.}}\{n^{1/2-\lambda_\delta}\}, \quad \lambda_\delta = \frac{\delta^2}{22(\delta+2)^2}.$$

[Dedecker, Merlevede and Rio \(2014\)](#) extend this line to empirical distribution functions of absolutely regular sequences in $\mathbb{R}^d$. These results give a polynomial improvement over the $\sqrt{n}$ scale, but their assumptions are formulated through β -type or absolute-regularity coefficients. Such coefficients are powerful when available, but they are not imposed through the causal data-generating recursion itself. For many nonlinear recursions and scale-type models, physical-dependence bounds can be verified directly from the innovation-driven system, whereas absolute-regularity verification may require additional Markov or smoothing structure.

The present paper addresses the strong sequential lower-orthant approximation under a stationary causal representation. The assumptions are imposed through physical dependence of $X_i = g(\xi_i)$ and through local boundary regularity of lower orthants, rather than through S -mixing, strong mixing, or absolute regularity. This change of dependence language is substantive: it requires transferring physical dependence from the original process to the discontinuous indicator class, controlling local boundary slabs over $\mathbb{R}^d$, and preserving the sequential covariance $n(u \wedge v)\Gamma(s, t)$ after finite-window truncation and blockwise coupling. The resulting approximation gives an explicit polynomial rate below the $\sqrt{n}$ empirical-process scale and covers model classes whose verification proceeds from the causal recursion itself.

The causal representation is

$$X_i = g(\xi_i), \quad \xi_i = (\dots, \varepsilon_{i-1}, \varepsilon_i),$$

where the innovations are iid. Let $X_k^{*(0)}$ denote the coupled copy of X_k obtained by replacing only the innovation ε_0 in the causal input by an independent copy. The main technical difficulty is that the map $s \mapsto \mathbf{1}(X_i \leq s)$ is discontinuous. Replacing a remote innovation changes the indicator only when X_i crosses a lower-orthant boundary. Under bounded marginal densities and moment control, this gives the inheritance bound

$$\sup_s \left\| \mathbf{1}(X_k \leq s) - \mathbf{1}(X_k^{*(0)} \leq s) \right\|_r \lesssim \left\| |X_k - X_k^{*(0)}|_\infty \right\|_q^{q/\{r(q+1)\}}, \quad r \geq 1,$$

which transfers causal physical dependence to the dependent indicator class. The continuum in s is then handled by a local lower-orthant oscillation theorem. For a perturbation radius $b_n \downarrow 0$, define

$$\mathcal{D}_n(b_n) = \sup_{\substack{s, t \in \mathbb{R}^d \\ |s-t|_\infty \leq b_n}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n [\mathbf{1}(X_i \leq s) - \mathbf{1}(X_i \leq t) - \{F(s) - F(t)\}] \right|.$$

The local oscillation theorem proves, under the boundary conditions stated below,

$$\mathcal{D}_n(b_n) = O_{\mathbb{P}}(\sqrt{b_n \log n}) + O_{\mathbb{P}}(b_n L_n^\partial).$$

Here L_n^∂ is a deterministic bound for the predictable boundary-density process and is $O(\sqrt{\log n})$ in the model classes verified later. The first term is the martingale fluctuation over local coordinate slabs, and the second term is the predictable boundary contribution. This result is the empirical-process input that carries finite-grid Gaussian couplings to the full index set $\mathbb{R}^d$. It complements the local oscillation bounds of [Wu \(2006\)](#) and the concentration inequalities for linear time-series empirical processes of [Chen and Wu \(2018\)](#).

The main result is the sequential coupling. If $p > d$ is the available moment exponent, define

$$\gamma_0 = \frac{p + d(p+1)}{3p + d(2p+1)} < \frac{1}{2}, \quad \gamma_{\text{seq}} = \frac{5}{4(3 - \gamma_0)} < \frac{1}{2}.$$

For every $\eta > 0$, the endpoint result constructs $T_n(1, \cdot)$ with covariance $n\Gamma(\cdot, \cdot)$ and rate

$$\sup_{s \in \mathbb{R}^d} |S_n(1, s) - T_n(1, s)| = O_{\mathbb{P}}(n^{\gamma_0 + \eta}).$$

Under geometric physical dependence and the same lower-orthant boundary conditions, for every $\eta > 0$ the full sequential Gaussian process in (1.2) can be constructed so that

$$\sup_{u \in [0,1]} \sup_{s \in \mathbb{R}^d} |S_n(u, s) - T_n(u, s)| = O_{\mathbb{P}}(n^{\gamma_{\text{seq}} + \eta}).$$

The endpoint exponent γ_0 is obtained by balancing finite-grid martingale coupling with lower-orthant oscillation. The sequential exponent γ_{seq} is the cost of making the approximation uniform in u , through independent-tail finite-window truncation, big-small blocking, blockwise coupling, and Gaussian stitching. The same proof scheme also covers polynomial physical-dependence decay. If

$$\beta = \frac{(\alpha - 1)q}{q + 1}, \quad \beta_{\text{crit}}(p, d) = \frac{p(16d + 25) + 7d}{2(p - d)},$$

then the polynomial theorem applies under $\beta > \beta_{\text{crit}}(p, d)$, equivalently $\alpha > 1 + \{(q + 1)/q\}\beta_{\text{crit}}(p, d)$.

The proof has five main steps. First, the indicator-inheritance bound converts physical dependence of X_i into dependence bounds for the discontinuous indicator class. Second, the index set is localized and discretized on a tail-adapted lower-orthant grid. Third, a finite-dimensional martingale coupling is applied on the grid. Fourth, the endpoint coupling is transferred to all thresholds by the local oscillation theorem. Fifth, the sequential process is obtained by independent-tail block approximation and a Gaussian stitching argument that restores the covariance $n(u \wedge v)\Gamma(s, t)$.

The approximation yields several consequences. The sequential bridge

$$\mathbb{D}_n(u, s) = n^{-1/2}\{S_n(u, s) - uS_n(1, s)\}$$

is approximated by the Gaussian bridge

$$\mathbb{B}_n(u, s) = n^{-1/2}\{T_n(u, s) - uT_n(1, s)\},$$

whose covariance is $(u \wedge v - uv)\Gamma(s, t)$. This supplies a direct Gaussian calibration for distributional change-point statistics under serial dependence. The same coupling gives endpoint distributional and rectangle probability consequences, sequential scalar quantile approximations uniformly over $u \in [u_0, 1]$, and a finite-grid dependent multiplier calibration whose conditional covariance is a Bartlett long-run covariance estimator. The latter connects the multiplier construction to HAC covariance estimation (Newey and West, 1987) and to dependent empirical-process bootstrap methods (Buehlmann, 1994; Buecher and Kojadinovic, 2016). Financial monitoring provides one application. Changes in the joint distribution of returns, volatility, and funding or credit stress can carry information beyond changes in individual coordinates, and the COVID-19 period offers a setting in which equity prices and volatility moved sharply while financial channels amplified the real shock (Baker et al., 2020; Ramelli and Wagner, 2020).

The main contributions are as follows.

- *Sequential lower-orthant coupling.* We construct a covariance-matched Gaussian process for the empirical distribution process of stationary causal time series. In the geometric case, the coupling satisfies $O_{\mathbb{P}}(n^{\gamma_{\text{seq}} + \eta}) = o_{\mathbb{P}}(\sqrt{n})$, and the polynomial case is covered under an explicit polynomial-decay condition.
- *Empirical-process input for discontinuous indicators.* We prove a local lower-orthant oscillation inequality that separates the independent-type fluctuation over boundary slabs from a predictable boundary contribution.
- *Distributional inference consequences.* We derive bridge, change-point, rectangle probability, scalar quantile, and finite-grid multiplier results from the same sequential coupling, with model-level verification for additive, linear, contractive nonlinear, and scale-type causal processes.

Organization. Section 2 gives the setup, assumptions, and main Gaussian approximation theorems. Section 3 records distributional, bridge, change-point, and quantile consequences. Section 4 gives the finite-grid dependent multiplier calibration theorem. Section 5 verifies the assumptions for representative causal models. Section 6 reports a simulation study and a real data analysis. Section 7 discusses theoretical scope, interpretation, and extensions. The supplementary material contains the empirical-process inputs, the sequential block proof, and proofs of the statistical consequences.

2. Setup, assumptions, and main results. This section states the assumptions and the approximation theorems. We first give the boundary and oscillation tools that pass from finite grids to the continuum of lower orthants. We then state the endpoint, geometric sequential, and polynomial sequential Gaussian approximations.

Let d be fixed. The process $(X_i)_{i \in \mathbb{Z}}$ takes values in $\mathbb{R}^d$ and has the causal representation

$$(2.1) \quad X_i = g(\xi_i), \quad \xi_i = (\dots, \varepsilon_{i-1}, \varepsilon_i),$$

where $(\varepsilon_i)_{i \in \mathbb{Z}}$ are iid. For $s \in \mathbb{R}^d$, set

$$(2.2) \quad f_s(x) = \mathbf{1}(x \leq s) - F(s), \quad F(s) = \mathbb{P}(X_0 \leq s).$$

The sequential empirical process is

$$(2.3) \quad S_n(u, s) = \sum_{i=1}^{\lfloor nu \rfloor} f_s(X_i), \quad (u, s) \in [0, 1] \times \mathbb{R}^d.$$

Let $X_k^{*(0)}$ be obtained from X_k by replacing only ε_0 in the causal input by an independent copy. For $q \geq 1$, define

$$(2.4) \quad \delta_q(k) = \left\| |X_k - X_k^{*(0)}|_{\infty} \right\|_q.$$

2.1. Notation and conventions. For $s \in \mathbb{R}^d$, inequalities such as $x \leq s$ are coordinatewise and $A_s = (-\infty, s_1] \times \dots \times (-\infty, s_d]$. For $A \geq 1$, write $K_A = [-A, A]^d$. The norm $|x|_{\infty}$ is the coordinate sup norm, and $\|Z\|_r = (\mathbb{E}|Z|^r)^{1/r}$. For nonnegative deterministic sequences, $a_n \lesssim b_n$ means $a_n \leq Cb_n$ for a constant $C < \infty$ not depending on n , and $a_n \asymp b_n$ means both $a_n \lesssim b_n$ and $b_n \lesssim a_n$. The symbol $\log^C n$ denotes a possibly changing finite power of $\log n$. Constants $C, c, C_0, \dots$ may change from line to line, but never depend on n, u, s, t unless this is explicitly stated.

All suprema over lower orthants are interpreted through a fixed countable rational-endpoint subclass on localization cubes and then extended by monotone or L^2 -separability. The Gaussian processes constructed below are taken in separable versions. This convention

is used only to avoid measurability distractions; every stochastic bound is first proved on the countable subclass and then extended by the displayed oscillation estimates.

The sequential proof uses several length parameters. We reserve ℓ_n for big-block lengths, m_n for finite-window truncation lengths, and h_n for small-block lengths. Rate exponents, such as r_{poly} in Theorem 2.14, are kept separate from these length parameters.

REMARK 2.1 (Role of the assumptions). The assumptions below have three roles. Condition 2.2 localizes the lower-orthant class and controls marginal boundary slabs. Conditions 2.4 and 2.5 quantify serial dependence through the causal representation (2.1). Conditions 2.6–2.8 control the off-grid lower-orthant oscillation by requiring finite-step anti-concentration and a manageable predictable boundary-density process.

CONDITION 2.2 (Moment and marginal slab regularity). For some $p > d$, $\mathbb{E}|X_0|_\infty^p < \infty$. Each coordinate marginal distribution of X_0 admits a density, and these coordinate densities are uniformly bounded.

REMARK 2.3 (Moment localization and tail control). The moment condition is used to localize the full index set. If $K_A = [-A, A]^d$, then

$$\mathbb{P}(X_0 \notin K_A) \leq A^{-p} \mathbb{E}|X_0|_\infty^p,$$

so a cube with $A = A_n$ makes the tail contribution small uniformly over lower orthants. A mesh of width δ on K_A has order

$$M_\delta \asymp (A/\delta)^d$$

grid points. Thus moment localization and d -dimensional grid entropy enter the same rate calculation, producing the endpoint exponent γ_0 in (2.10). The restriction $p > d$ makes this exponent strictly smaller than $1/2$. For linear time series, Chen and Wu (2018) show that dependence and heavy tails create polynomial deviation terms even for bounded indicator-indexed empirical processes, and their half-interval bounds are sharp up to constants.

CONDITION 2.4 (Geometric physical dependence). For some $q \geq 1$, $C < \infty$, and $\rho \in (0, 1)$,

$$\delta_q(k) \leq C\rho^k, \quad k \geq 1.$$

CONDITION 2.5 (Polynomial physical dependence). For some $q \geq 1$, $C < \infty$, and $\alpha > 1$,

$$\delta_q(k) \leq C(1+k)^{-\alpha}, \quad k \geq 0.$$

The long-run covariance kernel is

$$(2.5) \quad \Gamma(s, t) = \sum_{h \in \mathbb{Z}} \text{Cov}\{f_s(X_0), f_t(X_h)\}.$$

Under the assumptions below the series is absolutely convergent. The sequential Gaussian target has covariance

$$(2.6) \quad \text{Cov}\{T_n(u, s), T_n(v, t)\} = n(u \wedge v)\Gamma(s, t).$$

When $u = 1$, the unnormalized endpoint covariance is $K_n(s, t) = n\Gamma(s, t)$. We reserve the notation Γ for the normalized long-run covariance throughout the sequential paper.

2.2. *Lower-orthant boundary conditions.* The off-grid step requires local control of lower-orthant boundary slabs. For $A \geq 1$, put $K_A = [-A, A]^d$. For $b_n \downarrow 0$, define

$$(2.7) \quad \mathcal{D}_n(b_n) = \sup_{\substack{s, t \in \mathbb{R}^d \\ |s-t|_\infty \leq b_n}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n [\mathbf{1}(X_i \leq s) - \mathbf{1}(X_i \leq t) - \{F(s) - F(t)\}] \right|.$$

CONDITION 2.6 (Finite-step slab anti-concentration). There exist an integer $L \geq 1$ and $C_0 < \infty$ such that, for every coordinate j , every $a \in \mathbb{R}$, and every $h > 0$,

$$\mathbb{P}(a < X_{L,j} \leq a + h \mid \xi_0) \leq C_0 h \quad \text{a.s.}$$

Let $F_L(s \mid \xi) = \mathbb{P}(X_L \leq s \mid \xi)$. Write the coordinate boundary derivative, when it exists, as

$$q_{L,j}(z, u_{-j} \mid \xi) = \frac{\partial}{\partial s_j} F_L(s \mid \xi) \Big|_{s_j = z, s_{-j} = u_{-j}}.$$

CONDITION 2.7 (Boundary differentiability). For the same L as in Condition 2.6, $F_L(\cdot \mid \xi)$ is coordinatewise absolutely continuous on compact rectangles a.s. The derivatives $q_{L,j}$ admit jointly measurable versions and satisfy the path identity

$$F_L(t \mid \xi) - F_L(s \mid \xi) = \sum_{j=1}^d \int_{s_j}^{t_j} q_{L,j}(z, v_{-j}^{(j)} \mid \xi) dz$$

for all s, t in the localization cubes used below. Here

$$v^{(0)} = s, \quad v^{(j)} = (t_1, \dots, t_j, s_{j+1}, \dots, s_d).$$

Set

$$\zeta_i^{(j)}(w) = q_{L,j}(w \mid \xi_{i-L}) - \mathbb{E} q_{L,j}(w \mid \xi_0), \quad w \in \mathbb{R}^d,$$

and

$$R_n^\partial(A) = \max_{1 \leq j \leq d} \sup_{w \in K_A} \left| n^{-1/2} \sum_{i=1}^n \zeta_i^{(j)}(w) \right|.$$

CONDITION 2.8 (Uniform local boundary predictability). For every deterministic sequence $b_n \downarrow 0$ satisfying $\log n = O(nb_n)$, define the localization radius

$$(2.8) \quad A_n(b_n) = 1 \vee \left\{ \frac{n(\mathbb{E}|X_0|_\infty^p)^2}{b_n \log n} \right\}^{1/(2p)},$$

and suppose there exists a deterministic sequence $L_n^\partial = n^{o(1)}$, possibly depending on the chosen sequence b_n , such that

$$R_n^\partial\{A_n(b_n)\} = O_\mathbb{P}(L_n^\partial).$$

In the model classes in Section 5, one may take $L_n^\partial = O(\sqrt{\log n})$.

THEOREM 2.9 (Local lower-orthant oscillation). Suppose $b_n \rightarrow 0$ and $\log n = O(nb_n)$. Under Conditions 2.2, 2.6, 2.7, and 2.8,

$$(2.9) \quad \mathcal{D}_n(b_n) = O_\mathbb{P}(\sqrt{b_n \log n}) + O_\mathbb{P}(b_n L_n^\partial).$$

REMARK 2.10 (Boundary slabs versus scalar endpoints). For $d = 1$, moving from s to t changes a half-line by one interval. For $d > 1$, the symmetric difference $A_s \triangle A_t$ is contained in a union of d coordinate slabs whose conditional probabilities depend on the predictive boundary densities $q_{L,j}$. Conditions 2.6– 2.8 are precisely the assumptions needed to turn this geometric boundary decomposition into the stochastic oscillation bound (2.9). The first term $\sqrt{b_n \log n}$ is the martingale fluctuation over slabs, and the second term $b_n L_n^\partial$ is the predictable boundary contribution.

2.3. *Gaussian approximation theorems.* We now state the Gaussian approximation results. The endpoint theorem gives the finite-time empirical distribution coupling, and the sequential theorem constructs one Gaussian process T_n on $[0, 1] \times \mathbb{R}^d$ with covariance (2.6). The exponents below quantify the approximation error; the object being approximated is the process S_n in (2.3).

Define

$$(2.10) \quad \gamma_0 = \frac{p + d(p + 1)}{3p + d(2p + 1)} < \frac{1}{2}$$

and

$$(2.11) \quad \gamma_{\text{seq}} = \frac{5}{4(3 - \gamma_0)} < \frac{1}{2}.$$

Equivalently, if $a = 2\gamma_{\text{seq}}$, then

$$(2.12) \quad \frac{5}{2} - a \left(\frac{5}{2} - \gamma_0 \right) = \frac{a}{2} = \gamma_{\text{seq}}.$$

The endpoint exponent γ_0 comes from the balance between finite-grid martingale coupling and lower-orthant oscillation. The sequential exponent γ_{seq} records the additional powers introduced by truncation, block accumulation, and Gaussian stitching.

REMARK 2.11 (Limiting rate behavior). The endpoint exponent deteriorates to the empirical-process boundary as the available moment approaches the localization threshold:

$$\lim_{p \downarrow d} \gamma_0 = \frac{1}{2}.$$

With arbitrarily high moments,

$$\lim_{p \rightarrow \infty} \gamma_0 = \frac{d + 1}{2d + 3}, \quad \lim_{p \rightarrow \infty} \gamma_{\text{seq}} = \frac{5(2d + 3)}{4(5d + 8)}.$$

For $d = 1$, these limits are $2/5$ and $25/52$, respectively. The gap between γ_0 and γ_{seq} records the price of uniformity in the time index.

THEOREM 2.12 (Endpoint lower-orthant Gaussian approximation). *Assume Conditions 2.2, 2.4, 2.6, 2.7, and 2.8. Then, for every $\eta > 0$, one can construct a centered Gaussian process $T_n(1, \cdot)$ with covariance $K_n(s, t) = n\Gamma(s, t)$ such that*

$$(2.13) \quad \sup_{s \in \mathbb{R}^d} |S_n(1, s) - T_n(1, s)| = O_{\mathbb{P}}(n^{\gamma_0 + \eta}).$$

THEOREM 2.13 (Geometric sequential Gaussian approximation). *Assume the conditions of Theorem 2.12. Then, for every $\eta > 0$, one can construct a centered Gaussian process $\{T_n(u, s) : (u, s) \in [0, 1] \times \mathbb{R}^d\}$ with covariance (2.6) such that*

$$(2.14) \quad \sup_{u \in [0, 1]} \sup_{s \in \mathbb{R}^d} |S_n(u, s) - T_n(u, s)| = O_{\mathbb{P}}(n^{\gamma_{\text{seq}} + \eta}).$$

The proof of Theorem 2.13 constructs independent-tail finite-window truncations, proves a terminal coupling for the resulting m -dependent blocks, and then stitches the Gaussian blocks into the process with covariance (2.6). The resulting covariance has the nested sequential clock $n(u \wedge v)\Gamma(s, t)$, which is not obtained by applying Theorem 2.12 separately at finitely many time points.

For polynomial decay, put

$$(2.15) \quad \beta = \frac{(\alpha - 1)q}{q + 1}, \quad \lambda_H = \frac{\beta}{2} - 1.$$

Set also

$$(2.16) \quad \beta_{\text{crit}}(p, d) = \frac{9 - 2\gamma_0}{2(1 - 2\gamma_0)} = \frac{p(16d + 25) + 7d}{2(p - d)}.$$

THEOREM 2.14 (Polynomial sequential Gaussian approximation). *Assume Conditions 2.2, 2.5, 2.6, 2.7, and 2.8. Suppose, in addition, that*

$$(2.17) \quad \alpha > 1 + \frac{q + 1}{q} \beta_{\text{crit}}(p, d).$$

Equivalently, $\beta > \beta_{\text{crit}}(p, d)$. Then there exists $r_{\text{poly}} = r_{\text{poly}}(p, q, \alpha, d) < 1/2$ such that one can construct a centered Gaussian process T_n with covariance (2.6) satisfying, for every $\eta > 0$,

$$(2.18) \quad \sup_{u \in [0, 1]} \sup_{s \in \mathbb{R}^d} |S_n(u, s) - T_n(u, s)| = O_{\mathbb{P}}(n^{r_{\text{poly}} + \eta}).$$

In particular the approximation error is $o_{\mathbb{P}}(\sqrt{n})$.

REMARK 2.15 (Admissible polynomial rates). The condition in Theorem 2.14 gives a nonempty range of rates. Indeed, put

$$a^\circ = \frac{8}{9 - 2\gamma_0}, \quad u^\circ = 1 - a^\circ = \frac{1 - 2\gamma_0}{9 - 2\gamma_0}.$$

Since $\beta_{\text{crit}}(p, d) = 1/(2u^\circ)$, the condition $\beta > \beta_{\text{crit}}(p, d)$ allows one to choose the block exponents a and b , corresponding to $\ell_n = \lfloor n^a \rfloor$ and $m_n = h_n = \lfloor n^b \rfloor$ in the proof, so that

$$\frac{1}{2\beta} < b < \min \left\{ u^\circ, \frac{1}{\beta} \right\}, \quad a = a^\circ.$$

Lemma F.1 shows that this choice makes the block-coupling, truncation, small-block, covariance-residual, and Gaussian-modulus exponents smaller than $1/2$. Hence r_{poly} may be chosen between their maximum and $1/2$. In the limit $p, q \rightarrow \infty$, the sufficient threshold becomes $\beta > 8d + 25/2$, or equivalently $\alpha > 8d + 27/2$. For $d = 1$, this gives $\beta > 41/2$ and, when q is large, $\alpha > 43/2$. Here β is the decay exponent in (2.15), not a mixing coefficient.

2.4. Main proof ingredients. The proofs of Theorems 2.12, 2.13 and 2.14 use finite-grid coupling, local lower-orthant oscillation, independent-tail truncation, block coupling, and Gaussian stitching.

- (1) *Indicator and boundary reduction.* An indicator-inheritance bound converts physical dependence of the causal process into dependence bounds for the discontinuous indicators. Theorem 2.9 then controls the continuum error created when a finite lower-orthant grid is enlarged to all of $\mathbb{R}^d$. These two inputs isolate the effect of boundary crossings.

- (2) *End-point finite-grid coupling.* The proof of Theorem 2.12 localizes the index set, discretizes it on a tail-adapted grid, applies a martingale Yurinskii coupling on that grid, and uses empirical and Gaussian off-grid moduli. Balancing these terms gives the end-point exponent γ_0 .
- (3) *Independent-tail block approximation.* The sequential proof first replaces X_i by an independent-tail finite-window truncation, partitions time into big and small blocks, and applies the terminal lower-orthant coupling to the independent big blocks. This step supplies the block-level input for the sequential construction.
- (4) *Gaussian stitching.* The block Gaussian variables initially have a reduced time clock and the finite-window covariance Γ_m . The stitching step embeds them into a sequential Gaussian process with covariance $n(u \wedge v)\Gamma$, repairing both the clock and the covariance.
- (5) *Rate balance.* In the geometric case, truncation, block coupling, deleted-clock control, and stitching are balanced to obtain γ_{seq} . In the polynomial case, the same decomposition is retained, and the decay condition (2.17) permits block lengths for which every error term is $o_{\mathbb{P}}(\sqrt{n})$.

3. Consequences for distributional inference. The sequential approximation converts empirical distribution functionals into Gaussian functionals uniformly in time. This section states the consequences directly under the time-series assumptions of Section 2. For the geometric theorem write $\theta = \gamma_{\text{seq}}$; for the polynomial theorem write $\theta = r_{\text{poly}}$, where $r_{\text{poly}} < 1/2$ is the exponent supplied by Theorem 2.14. Throughout this section,

$$F_n(s) = n^{-1} \sum_{i=1}^n \mathbf{1}(X_i \leq s)$$

denotes the empirical distribution function based on the full sample.

3.1. *Endpoint distributional functionals.* Under Theorem 2.12, for every $\eta \in (0, 1/2 - \gamma_0)$,

$$(3.1) \quad \sup_{s \in \mathbb{R}^d} |\sqrt{n}\{F_n(s) - F(s)\} - Z_n(s)| = o_{\mathbb{P}}(1), \quad Z_n(s) = n^{-1/2}T_n(1, s).$$

Thus every statistic that is Lipschitz with respect to the supremum norm on the lower-orthant empirical process can be replaced by the corresponding functional of Z_n . Examples include simultaneous bands

$$\sup_{s \in \mathcal{S}_0} |\sqrt{n}\{F_n(s) - F(s)\}|,$$

Kolmogorov–Smirnov type statistics, and integrated-square statistics on a finite or compact computational set $\mathcal{S}_0$ for which the Gaussian functional is well defined. The covariance is the long-run covariance Γ , not the iid bridge covariance.

The lower-orthant result also gives rectangle probability inference by finite inclusion–exclusion. For a half-open rectangle $R = (a, b]$,

$$(3.2) \quad \mathbb{P}(X_0 \in R) = \sum_{\epsilon \in \{0,1\}^d} (-1)^{|\epsilon|} F(c_\epsilon), \quad c_{\epsilon,j} = \begin{cases} a_j, & \epsilon_j = 1, \\ b_j, & \epsilon_j = 0. \end{cases}$$

Consequently any fixed finite family of rectangle probabilities has a joint Gaussian approximation obtained by applying the same linear map to $\{Z_n(c_\epsilon)\}$. This gives the basic Gaussian input for joint-tail probability intervals and fixed-grid empirical copula diagnostics.

3.2. *Sequential bridge and change-point statistics.* Define the empirical bridge and its Gaussian analogue by

$$(3.3) \quad \mathbb{D}_n(u, s) = n^{-1/2}\{S_n(u, s) - uS_n(1, s)\}, \quad \mathbb{B}_n(u, s) = n^{-1/2}\{T_n(u, s) - uT_n(1, s)\}.$$

COROLLARY 3.1 (Sequential bridge approximation). *Assume either the geometric conditions of Theorem 2.13, with $\theta = \gamma_{\text{seq}}$, or the polynomial conditions of Theorem 2.14, with $\theta = r_{\text{poly}}$. Then for every $\eta \in (0, 1/2 - \theta)$ one can construct $\mathbb{B}_n$ such that*

$$(3.4) \quad \sup_{u \in [0, 1]} \sup_{s \in \mathbb{R}^d} |\mathbb{D}_n(u, s) - \mathbb{B}_n(u, s)| = O_{\mathbb{P}}(n^{\theta-1/2+\eta}) = o_{\mathbb{P}}(1).$$

Moreover $\mathbb{B}_n$ is centered Gaussian with covariance

$$(3.5) \quad \text{Cov}\{\mathbb{B}_n(u, s), \mathbb{B}_n(v, t)\} = (u \wedge v - uv)\Gamma(s, t).$$

For a compact or finite computational index set $\mathcal{S}_0 \subset \mathbb{R}^d$ and trimming constant $u_0 \in (0, 1/2)$, the distributional change-point statistic

$$(3.6) \quad CP_n = \sup_{u \in [u_0, 1-u_0]} \sup_{s \in \mathcal{S}_0} |\mathbb{D}_n(u, s)|$$

satisfies

$$(3.7) \quad CP_n = \sup_{u \in [u_0, 1-u_0]} \sup_{s \in \mathcal{S}_0} |\mathbb{B}_n(u, s)| + o_{\mathbb{P}}(1)$$

under the same geometric or polynomial assumptions. The limiting calibration is nuisance-dependent through Γ . Section 4 therefore develops a finite-grid dependent multiplier procedure that estimates this covariance through a Bartlett long-run covariance matrix.

COROLLARY 3.2 (Gaussian calibration of the change-point statistic). *Let*

$$G_n^{\text{cp}} = \sup_{u \in [u_0, 1-u_0]} \sup_{s \in \mathcal{S}_0} |\mathbb{B}_n(u, s)|.$$

Assume the conditions of Corollary 3.1. Let $c_{n,1-\alpha}$ be a Gaussian critical value satisfying

$$\mathbb{P}(G_n^{\text{cp}} \leq c_{n,1-\alpha}) = 1 - \alpha + o(1),$$

and suppose G_n^{cp} is asymptotically continuous at $c_{n,1-\alpha}$, in the sense that

$$\mathbb{P}(|G_n^{\text{cp}} - c_{n,1-\alpha}| \leq \delta_n) \rightarrow 0 \quad \text{for every deterministic } \delta_n \downarrow 0.$$

Then

$$(3.8) \quad \mathbb{P}(CP_n \leq c_{n,1-\alpha}) = 1 - \alpha + o(1).$$

REMARK 3.3 (Fixed distributional alternatives). The theory above is a null approximation for stationary data, but it also explains the deterministic drift targeted by bridge statistics. If a single change occurs at fraction $\tau_0 \in (0, 1)$, with marginal distribution F_1 before the change and F_2 after the change, then the empirical bridge has mean of order

$$\sqrt{n}(u \wedge \tau_0 - u\tau_0)\{F_1(s) - F_2(s)\}.$$

Thus any fixed alternative satisfying $\sup_s |F_1(s) - F_2(s)| > 0$ creates a drift of order $\sqrt{n}$, while the null Gaussian bridge remains $O_{\mathbb{P}}(1)$ after normalization.

REMARK 3.4 (Bridge covariance under dependence). If the observations were iid, $\Gamma(s, t)$ would reduce to $F(s \wedge t) - F(s)F(t)$. Under serial dependence it is instead

$$\Gamma(s, t) = \sum_{h \in \mathbb{Z}} \text{Cov}\{\mathbf{1}(X_0 \leq s), \mathbf{1}(X_h \leq t)\},$$

so positive dependence among indicators inflates the bridge variance. This is the reason iid Brownian-bridge critical values can be misleading for distributional stability tests with time-series data.

3.3. *Sequential scalar quantiles.* When $d = 1$, the sequential empirical process gives recursive quantile approximations. Let $\mathcal{T} = [\tau_L, \tau_U] \subset (0, 1)$, $q_\tau = F^{-1}(\tau)$, and $k_u = \lfloor nu \rfloor$. Define

$$F_{k_u}(x) = k_u^{-1} \sum_{i=1}^{k_u} \mathbf{1}(X_i \leq x), \quad \hat{q}_{u,\tau} = F_{k_u}^{-1}(\tau).$$

LEMMA 3.5 (Canonical metric of the scalar Gaussian target). Assume $d = 1$ and either the geometric conditions of Theorem 2.13 or the polynomial conditions of Theorem 2.14. Let T_n be the corresponding Gaussian process and put $Z_n(u, x) = n^{-1/2}T_n(u, x)$. Then there exist $C < \infty$ and $\kappa > 0$ such that, for all $u, v \in [0, 1]$ and $x, y \in \mathbb{R}$,

$$\{\mathbb{E}|Z_n(u, x) - Z_n(v, y)|^2\}^{1/2} \leq C\{|u - v|^{1/2} + |F(x) - F(y)|^\kappa\}.$$

LEMMA 3.6 (Gaussian modulus sufficient for sequential quantiles). Let Z_n be a separable centered Gaussian process on $[0, 1] \times \mathbb{R}$. Assume F is Lipschitz on an open interval containing $\{q_\tau : \tau \in \mathcal{T}\}$. Suppose that the canonical metric of Z_n satisfies, for all $u, v \in [0, 1]$ and $x, y \in \mathbb{R}$, and for some $C < \infty$ and $\kappa > 0$,

$$\{\mathbb{E}|Z_n(u, x) - Z_n(v, y)|^2\}^{1/2} \leq C\{|u - v|^{1/2} + |F(x) - F(y)|^\kappa\}.$$

Then $\sup_{u,x} |Z_n(u, x)| = O_{\mathbb{P}}(1)$, and for each fixed $M < \infty$,

$$(3.9) \quad \sup_{u \in [0,1]} \sup_{\tau \in \mathcal{T}} \sup_{|x - q_\tau| \leq M/\sqrt{n}} |Z_n(u, x) - Z_n(u, q_\tau)| = o_{\mathbb{P}}(1).$$

COROLLARY 3.7 (Scalar sequential quantile process). Assume $d = 1$ and either the geometric conditions of Theorem 2.13, with $\theta = \gamma_{\text{seq}}$, or the polynomial conditions of Theorem 2.14, with $\theta = r_{\text{poly}}$. Let T_n be the Gaussian process constructed in the corresponding theorem and put $Z_n(u, x) = n^{-1/2}T_n(u, x)$. Assume that F has a Lipschitz density f on an open interval containing $\{q_\tau : \tau \in \mathcal{T}\}$, and

$$0 < c_f \leq \inf_{\tau \in \mathcal{T}} f(q_\tau) \leq \sup_{\tau \in \mathcal{T}} f(q_\tau) \leq C_f < \infty.$$

Then, for any fixed $u_0 > 0$,

$$(3.10) \quad \sup_{u \in [u_0, 1]} \sup_{\tau \in \mathcal{T}} \left| \frac{k_u}{\sqrt{n}} (\hat{q}_{u,\tau} - q_\tau) + \frac{Z_n(u, q_\tau)}{f(q_\tau)} \right| = o_{\mathbb{P}}(1).$$

REMARK 3.8 (Quantile applications). The factor $k_u/\sqrt{n}$ is the natural normalization for recursive quantiles because the partial sample size is k_u while the Gaussian process is normalized by $\sqrt{n}$. The corollary applies to fixed marginal quantiles of a stationary loss series. In dynamic value-at-risk or expected-shortfall models (Engle and Manganelli, 2004; Patton, Ziegel and Chen, 2019; Bayer and Dimitriadis, 2022; Horvath et al., 2022), the same empirical-process term appears after the model-specific linearization of residual indicators or quantile scores.

4. Finite-grid dependent multiplier calibration. The continuum Gaussian process in Theorems 2.13 and 2.14 is an approximation target. Feasible calibration is implemented on a finite grid. Let

$$\Theta_n = \{s_1, \dots, s_{M_n}\} \subset \mathbb{R}^d, \quad \mathcal{U}_n = \{u_1, \dots, u_{K_n}\} \subset [u_0, 1 - u_0],$$

where $u_0 \in (0, 1/2)$ and $M_n K_n \leq n^C$. Put

$$Y_i(j) = \mathbf{1}(X_i \leq s_j) - F(s_j), \quad \bar{Y}_n(j) = n^{-1} \sum_{i=1}^n Y_i(j), \quad \hat{Y}_i(j) = Y_i(j) - \bar{Y}_n(j).$$

Equivalently, $\hat{Y}_i(j) = \mathbf{1}(X_i \leq s_j) - F_n(s_j)$, so the procedure does not require knowledge of $F(s_j)$. The bandwidth $b_{\text{mul},n} \asymp n^{1/4}$ below is a theoretical default used to make the covariance proof transparent; data-driven bandwidth selection is a separate issue. Let $b_{\text{mul}} = b_{\text{mul},n}$ be a Bartlett bandwidth and let $e_i \stackrel{\text{iid}}{\sim} N(0, 1)$, independent of the data. Define

$$\omega_i = \frac{1}{\sqrt{b_{\text{mul}} + 1}} \sum_{\ell=0}^{b_{\text{mul}}} e_{i-\ell}, \quad G_n^*(u, j) = \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nu \rfloor} \omega_i \hat{Y}_i(j),$$

and

$$(4.1) \quad B_n^*(u, j) = G_n^*(u, j) - u G_n^*(1, j).$$

Conditionally on the observations, G_n^* is Gaussian. Its covariance is the sequential analogue of a Bartlett HAC estimator of the long-run covariance Γ .

THEOREM 4.1 (Finite-grid sequential multiplier validity). *Assume Conditions 2.2 and 2.4. By Lemma A.1, the grid indicators satisfy, for every fixed $v \geq 2$,*

$$\max_{j \leq M_n} \left\| Y_k(j) - Y_k^{*(0)}(j) \right\|_v \leq C_v \rho_v^k$$

for some $\rho_v \in (0, 1)$, uniformly in n , where $Y_k^{*(0)}(j)$ is formed by replacing ε_0 in the causal input. Suppose $M_n K_n \leq n^C$, $b_{\text{mul},n} \asymp n^{1/4}$, and

$$\min_{1 \leq j \leq M_n} \Gamma(s_j, s_j) \geq c_\Gamma > 0.$$

Let $B_n(u, j)$ be a centered Gaussian bridge with covariance

$$\text{Cov}\{B_n(u, j), B_n(v, k)\} = (u \wedge v - uv) \Gamma(s_j, s_k).$$

Then

$$(4.2) \quad \sup_{x \in \mathbb{R}} \left| \mathbb{P}^* \left(\max_{u \in \mathcal{U}_n} \max_{j \leq M_n} |B_n^*(u, j)| \leq x \right) - \mathbb{P} \left(\max_{u \in \mathcal{U}_n} \max_{j \leq M_n} |B_n(u, j)| \leq x \right) \right| \rightarrow_{\mathbb{P}} 0.$$

REMARK 4.2 (Scope of the finite-grid theorem). The finite-grid formulation matches the statistic used in simulations and applications. A continuum conditional multiplier theorem over $[0, 1] \times \mathbb{R}^d$ would require conditional oscillation theory for the multiplier process, data-dependent entropy bounds, and continuum Gaussian anti-concentration. The finite-grid result isolates the covariance-estimation step used for change-point and band calibration.

5. Model-level verification. The assumptions are stated through the causal mechanism rather than through mixing coefficients. Under the physical-dependence framework of Wu (2005b, 2008), verification proceeds by perturbing one innovation and tracing its effect through the data-generating recursion. This section records representative verification routes. The first proposition gives a general criterion for the boundary-predictability condition; the remaining propositions verify the physical-dependence assumptions for common causal models.

5.1. A predictive boundary criterion.

PROPOSITION 5.1 (Predictive boundary criterion). *Suppose that, for the finite prediction horizon L in Condition 2.6, there is a causal state H_i such that*

$$q_{L,j}(w \mid \xi_i) = Q_j(w, H_i), \quad j = 1, \dots, d.$$

Assume that Q_j is uniformly bounded and satisfies

$$|Q_j(w, h) - Q_j(w', h')| \leq C\{|w - w'|_\infty + d_{\mathcal{H}}(h, h')\}$$

on every localization cube K_A , with the same constant for $A \leq n^C$. Assume moreover that, for some fixed $v \geq 2$, the predictive state has summable physical dependence,

$$\sum_{k=0}^{\infty} \|d_{\mathcal{H}}(H_k, H_k^{*(0)})\|_v < \infty,$$

where $H_k^{(0)}$ is obtained by replacing ε_0 with an independent copy in the causal input. Then, for every deterministic localization radius $A_n \leq n^C$,*

$$R_n^\partial(A_n) = O_{\mathbb{P}}(\sqrt{\log n})$$

and Condition 2.8 holds uniformly over all deterministic $b_n \downarrow 0$ satisfying $\log n = O(nb_n)$, with $L_n^\partial = O(\sqrt{\log n})$.

5.2. Representative causal time-series models. We now combine Proposition 5.1 with standard physical-dependence calculations. The propositions are written as sufficient conditions: once the displayed state or scale dependence is verified, the abstract assumptions of Theorems 2.13 and 2.14 follow from the corresponding verification arguments.

PROPOSITION 5.2 (Additive finite-step model with smooth innovation density). *Suppose that, for the prediction horizon L ,*

$$X_{t+L} = H_t + \eta_{t,L},$$

where $\eta_{t,L}$ is independent of ξ_t , has density p_η on $\mathbb{R}^d$, and has finite p th moment. Assume that the coordinate marginal densities of $\eta_{t,L}$ are bounded and that, for each j ,

$$Q_j(w, h) = \int_{(-\infty, w_{-j} - h_{-j}]} p_\eta(w_j - h_j, z_{-j}) dz_{-j}$$

is uniformly bounded and Lipschitz in (w, h) on localization cubes K_A , with constants uniform for $A \leq n^C$. If, for some $q \geq 2$,

$$\left\| |H_k - H_k^{*(0)}|_\infty \right\|_q \leq C\rho^k \quad \text{or} \quad \left\| |H_k - H_k^{*(0)}|_\infty \right\|_q \leq C(1+k)^{-\alpha},$$

with $\alpha > 1$ in the polynomial case, then the corresponding geometric or polynomial physical-dependence condition holds for (X_t) . Consequently, the geometric or polynomial sequential theorem applies provided the stated moment condition in Condition 2.2 holds for X_t and, in the polynomial case, the decay exponent also satisfies (2.17).

PROPOSITION 5.3 (Additive finite-step and linear processes). *Let $X_t = G(H_t, \varepsilon_t)$, where G is Lipschitz in a predictive state H_t , and suppose the one-step or finite-step conditional distribution of X_t given H_{t-1} has boundary densities satisfying Proposition 5.1. If*

$$\left\| d_{\mathcal{H}}(H_k, H_k^{*(0)}) \right\|_q \leq C\rho^k \quad \text{or} \quad \left\| d_{\mathcal{H}}(H_k, H_k^{*(0)}) \right\|_q \leq C(1+k)^{-\alpha},$$

then Conditions 2.4 or 2.5, respectively, hold for X_t . In particular, stable linear processes

$$H_t = \sum_{\ell=0}^{\infty} A_{\ell} \zeta_{t-\ell}$$

are covered whenever $\sum_{\ell} \|A_{\ell}\|_{\infty \rightarrow \infty} < \infty$, the innovations have the required moments, and the coefficient tail is geometric or polynomial of the stated order. This includes vector linear filters and finite-order VARMA representations after the usual stable state-space embedding.

PROPOSITION 5.4 (Contractive nonlinear states). *Let $H_t = \Phi(H_{t-1}, \zeta_t)$, where*

$$d_{\mathcal{H}}\{\Phi(h, z), \Phi(h', z)\} \leq \rho d_{\mathcal{H}}(h, h'), \quad 0 < \rho < 1.$$

Assume the stationary solution has a finite q th moment in the state metric, and $X_t = G(H_t, \varepsilon_t)$ is Lipschitz in H_t . Then $\delta_q(k) \leq C\rho^k$. If the finite-step predictive distribution of X_t satisfies Conditions 2.6–2.8, the geometric sequential theorem applies.

REMARK 5.5 (Examples covered by the contractive form). The proposition covers Markovian iterated random functions with a uniform Lipschitz contraction, including many nonlinear autoregressions. When the conditional law of X_t given the predictive state has a bounded density and Lipschitz boundary derivatives, the slab and boundary conditions are checked by Proposition 5.1.

PROPOSITION 5.6 (Scalar and diagonal scale processes). *Let $X_{t,j} = \sigma_{t,j} \varepsilon_{t,j}$, $j = 1, \dots, d$, where ε_t has independent coordinates with bounded densities, bounded first derivatives, and $\sup_x |x f'_{\varepsilon_j}(x)| < \infty$, is independent of the past, and $\inf_{t,j} \sigma_{t,j} \geq c_{\sigma} > 0$ a.s. If*

$$\left\| |\sigma_k - \sigma_k^{*(0)}|_{\infty} \right\|_{2q} \leq C\rho^k \quad \text{or} \quad \left\| |\sigma_k - \sigma_k^{*(0)}|_{\infty} \right\|_{2q} \leq C(1+k)^{-\alpha},$$

and $\mathbb{E}|\sigma_0|^p + \mathbb{E}|\varepsilon_0|_{\infty}^p + \mathbb{E}|\varepsilon_0|_{\infty}^{2q} < \infty$, then the corresponding geometric or polynomial physical-dependence condition holds for X_t , and the finite-step slab and boundary-density conditions follow from the innovation densities.

6. Simulation and real data analysis. This section studies the bridge statistic and the finite-grid multiplier calibration in Section 4. The simulation experiment examines null size, power, and localization in a nonlinear causal model. The real data analysis applies the same finite-grid statistic to a short financial-stress data set around the COVID-19 period.

6.1. Simulation design. The simulated process is a three-dimensional smooth-threshold nonlinear autoregression,

$$(6.1) \quad X_t = AX_{t-1} + g_t b + \{B_0 + g_t B_1\} \tanh(X_{t-1}) + H(g_t) \varepsilon_t,$$

where $g_t \in (0, 1)$ is a logistic stress-state weight computed from X_{t-1} , and $H(g_t)H(g_t)^{\top}$ interpolates between low- and high-stress covariance matrices. The innovations are standardized Student t variables with ten degrees of freedom. Thus the design combines linear persistence, bounded nonlinear feedback, stress-dependent drift, and state-dependent volatility and correlation. The model provides a nonlinear dependent data-generating mechanism for the finite-grid bridge statistic.

All statistics are computed on a fixed lower-orthant grid of 125 points and on a trimmed time grid. For null calibration we simulate $R = 500$ independent paths at $n = 1000$ and $n = 3000$. For each path we compute bootstrap p -values using both the dependent multiplier

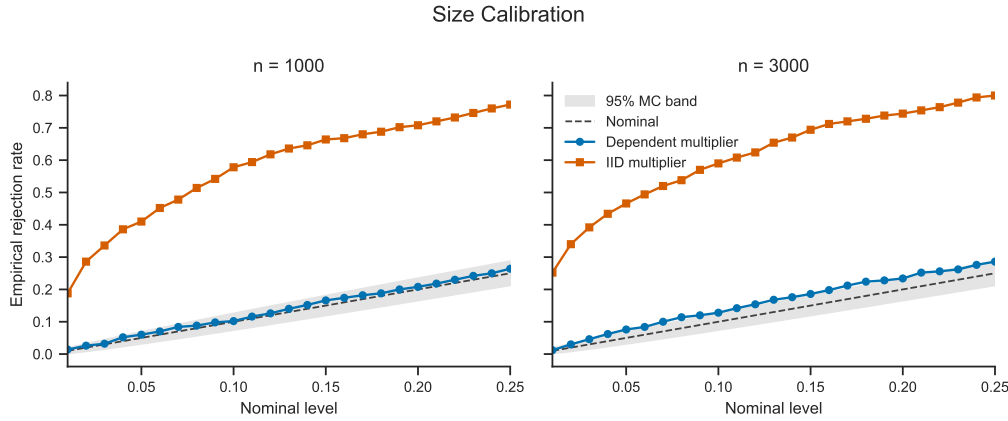


FIG 1. Null size calibration for the finite-grid bridge statistic. The grey band is the Monte Carlo uncertainty band around the nominal diagonal. The dependent multiplier tracks the nominal curve, whereas the iid multiplier substantially over-rejects under the dependent nonlinear data-generating process.

in Section 4 and an iid multiplier version that ignores serial dependence. The same stored p -values are evaluated over nominal levels $\alpha = 0.01, \dots, 0.25$. Each bootstrap calculation uses $B = 499$ multiplier draws.

For power and localization we introduce a single change at $\tau_0 = 0.5$. After the change point the stress-dependent drift is shifted from b to $b + \delta$, with $\delta \in \{0.10, 0.30\}$. This alternative changes the location of the process mainly in the stress state while leaving the rest of the mechanism fixed. Power is reported at $\alpha = 0.10$, using the dependent multiplier only, for $n \in \{500, 1000, 1500, 2000, 2500, 3000\}$. Localization is summarized by the median absolute error of the estimated change fraction.

6.2. Simulation results. Figure 1 reports the null size-calibration curves. The dependent multiplier curve tracks the nominal diagonal closely over the range of levels considered. At the commonly used level $\alpha = 0.10$, the empirical rejection rates are 0.102 for $n = 1000$ and 0.128 for $n = 3000$. The iid multiplier calibration is substantially more liberal: the corresponding rejection rates are 0.578 and 0.590. Thus, even for a finite-grid empirical bridge, the null distribution is governed by the long-run covariance of dependent indicators rather than by the iid Brownian-bridge covariance.

Figure 2 reports power and localization under the stress-state drift alternative. For the larger change, $\delta = 0.30$, power increases from 0.560 at $n = 500$ to 0.990 at $n = 3000$, and the median localization error decreases from 0.060 to 0.020. For the smaller change $\delta = 0.10$, power remains above the nominal level and is moderate, ranging from 0.218 to 0.372 across the sample sizes. Its localization error follows a more gradual pattern. These results support the use of the dependent multiplier bridge statistic for detectable stress-state distributional changes, with the largest power and localization gains under the larger change.

6.3. Real data analysis. The real data analysis applies the same statistic to daily U.S. financial data from FRED. The sample runs from 2019 to 2021 and uses the S&P 500 index, the CBOE VIX, and the TED spread (S&P Dow Jones Indices LLC, 2026; Chicago Board Options Exchange, 2026; Federal Reserve Bank of St. Louis, 2026). The three coordinates are the negative log return of the S&P 500, the log change of the VIX, and the first difference of the TED spread; each coordinate is then standardized over the sample. The TED spread provides a daily short-term funding-stress coordinate over the full sample; it is used in place

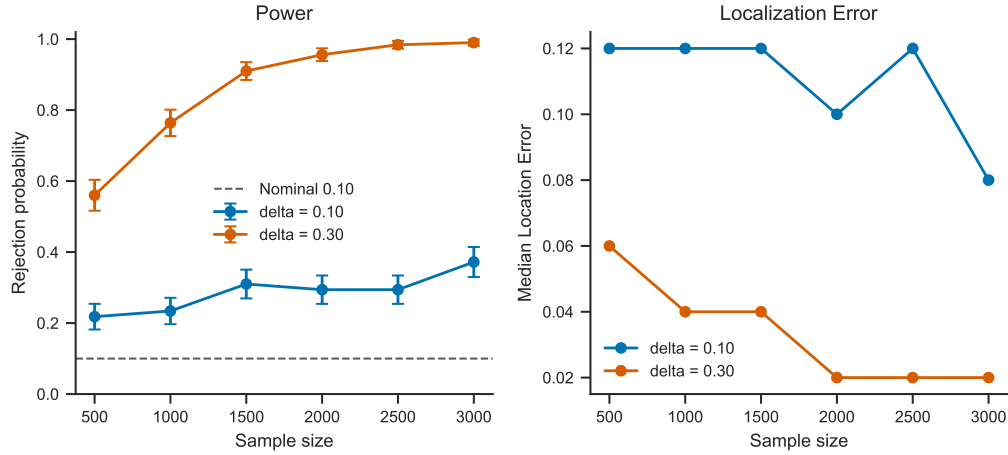


FIG 2. Power and localization under a stress-dependent drift change at $\tau_0 = 0.5$. The left panel reports rejection probabilities at $\alpha = 0.10$. The right panel reports the median absolute error of the estimated change fraction.

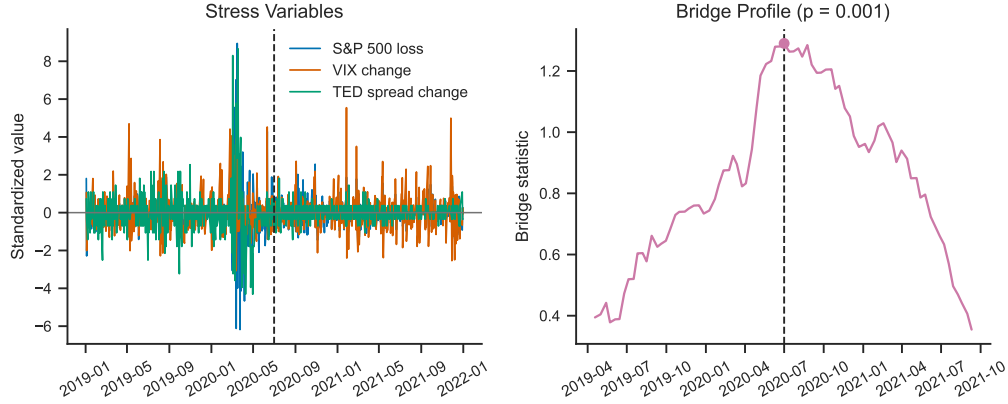


FIG 3. Real data analysis for daily U.S. financial stress variables, 2019–2021. The left panel shows the standardized stress vector. The right panel shows the finite-grid bridge profile computed from the same lower-orthant statistic; the dashed line marks the maximizing split date.

of the ICE high-yield option-adjusted spread series, whose current FRED release window starts after the COVID-19 period used here.

Figure 3 shows the standardized stress variables and the corresponding bridge profile. The dependent multiplier p -value is 0.001, and the maximum of the bridge profile occurs on July 1, 2020. We interpret this date as the sample split at which the cumulative distributional contrast in the standardized stress vector is largest. This interpretation is consistent with the role of COVID-19 as a major shock to U.S. equity and volatility markets documented in recent empirical finance work (Baker et al., 2020; Ramelli and Wagner, 2020). The analysis places the statistic in a setting where distributional instability is economically plausible, while the simulation study above provides the calibration evidence.

7. Discussion. The results establish a sequential Gaussian approximation for lower-orthant empirical distribution processes generated by stationary causal time series. The Gaussian target is indexed by both time and threshold and has covariance $n(u \wedge v)\Gamma(s, t)$, where Γ is the long-run covariance kernel of the dependent indicator process. This covariance structure determines the bridge, endpoint distributional, quantile, and finite-grid multiplier consequences developed above.

Two technical interfaces are central to the argument. First, the physical-dependence coefficients of the causal process are transferred to discontinuous lower-orthant indicators through boundary-crossing bounds. Second, local slab oscillation controls the error incurred when a finite lower-orthant grid is extended to the continuum threshold space. These inputs are combined with independent-tail truncation, big-small blocking, terminal block coupling, and Gaussian stitching to obtain the nested sequential clock $n(u \wedge v)\Gamma$.

The polynomial-dependence theorem uses the same construction under sufficiently fast polynomial physical-dependence decay. Its proof tracks the corresponding block choices together with the truncation, small-block, block-coupling, covariance-residual, and Gaussian-modulus terms. The finite-grid multiplier theorem complements this coupling result by giving a feasible conditional Gaussian calibration whose covariance is a Bartlett long-run covariance estimator for the lower-orthant indicator vector.

The numerical study is consistent with this covariance structure. In the nonlinear stress-threshold design, the dependent multiplier tracks the nominal size curve more closely than the iid multiplier. Under a stress-state location change, the bridge statistic has substantial power and localization gains for the larger change size. The real data analysis applies the same statistic to a COVID-era financial-stress vector and identifies the split at which the cumulative distributional contrast is largest.

Several extensions are natural within this framework. Adaptive grid selection would choose lower-orthant regions with high distributional sensitivity. Bandwidth selection for the multiplier covariance estimator can be linked to HAC theory. Local alternatives and post-selection inference for the estimated change point would connect the null Gaussian approximation with inference after a distributional change is detected.

Acknowledgments. Additional acknowledgments to be added.

Funding. Funding information to be added.

SUPPLEMENTARY MATERIAL

Supplement to “Sequential Gaussian Approximation for Empirical Distribution Processes of Dependent Causal Time Series”

Contains the proofs of the empirical-process inputs, the sequential block coupling, the statistical consequences, the finite-grid multiplier theorem, the model-verification results, and implementation details for the simulation and real data analysis.

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Supplement to “Sequential Gaussian Approximation for Empirical Distribution Processes of Dependent Causal Time Series”

Contains the proofs of the empirical-process inputs, the sequential block coupling, the statistical consequences, the finite-grid multiplier theorem, the model-verification results, and implementation details for the simulation and real data analysis.

APPENDIX A: END-POINT PROOF AND EMPIRICAL-PROCESS INPUTS

This appendix proves the end-point approximation theorem and the empirical-process inputs reused in the sequential argument. It is organized into four parts: indicator inheritance, local lower-orthant oscillation, finite-grid martingale coupling, and the final end-point grid/off-grid balance.

A.1. Indicator inheritance.

LEMMA A.1 (Lower-orthant indicator inheritance). *Assume Condition 2.2. Let X'_k be any coupled copy of X_k with the same marginal distribution, and let $u \geq 1$. Then*

$$(A.1) \quad \sup_{s \in \mathbb{R}^d} \|\mathbf{1}(X_k \leq s) - \mathbf{1}(X'_k \leq s)\|_u \leq C_u \|X_k - X'_k\|_\infty \|q^{q/\{u(q+1)\}}\|_q.$$

Consequently, under Condition 2.4 the indicator dependence is geometric, while under Condition 2.5 it is polynomial with exponent $\alpha q/\{u(q+1)\}$.

PROOF. Fix $s \in \mathbb{R}^d$ and $\eta > 0$. If the two indicators differ, then either $|X_k - X'_k|_\infty > \eta$, or for at least one coordinate j , $X_{k,j}$ lies in the slab $(s_j - \eta, s_j + \eta]$. Hence

$$(A.2) \quad \{\mathbf{1}(X_k \leq s) \neq \mathbf{1}(X'_k \leq s)\} \subset \{|X_k - X'_k|_\infty > \eta\} \cup \bigcup_{j=1}^d \{|X_{k,j} - s_j| \leq \eta\}.$$

By Markov's inequality and the uniformly bounded coordinate densities,

$$(A.3) \quad \mathbb{P}\{\mathbf{1}(X_k \leq s) \neq \mathbf{1}(X'_k \leq s)\} \leq \eta^{-q} \mathbb{E}|X_k - X'_k|_\infty^q + C\eta.$$

Choose

$$(A.4) \quad \eta = \|X_k - X'_k\|_\infty \|q^{q/(q+1)}\|_q.$$

Then (A.3) gives

$$\sup_{s \in \mathbb{R}^d} \mathbb{P}\{\mathbf{1}(X_k \leq s) \neq \mathbf{1}(X'_k \leq s)\} \leq C \|X_k - X'_k\|_\infty \|q^{q/(q+1)}\|_q.$$

Since the difference of two indicators has absolute value at most one, its L^u -norm is the u th root of the probability of disagreement. Combining this observation with (A.2)–(A.4) proves (A.1). $\square$

A.2. Proof of local lower-orthant oscillation.

LEMMA A.2 (Bounded-jump VC martingale inequality). *Fix $N_0 \geq 1$. Let $(\mathcal{G}_i)_{i \geq 0}$ be a filtration, let Z_i be $\mathcal{G}_i$ -measurable random elements in a standard Borel space, and let $\mathcal{C}$ be a countable measurable class with envelope $H \leq B$. Suppose that, for constants $A, V < \infty$,*

$$N\{\epsilon \|H\|_{Q,2}, \mathcal{C}, L^2(Q)\} \leq (A/\epsilon)^V, \quad 0 < \epsilon < 1,$$

for every probability measure Q . For $h \in \mathcal{C}$, set

$$M_i(h) = h(Z_i) - \mathbb{E}\{h(Z_i) \mid \mathcal{G}_{i-1}\}, \quad i = 1, \dots, N_0.$$

If, for some $v > 0$,

$$\sup_{h \in \mathcal{C}} \sum_{i=1}^{N_0} \mathbb{E}\{M_i(h)^2 \mid \mathcal{G}_{i-1}\} \leq v \quad a.s.,$$

then, for every $x \geq 1$,

$$(A.5) \quad \mathbb{P} \left[\sup_{h \in \mathcal{C}} \left| \sum_{i=1}^{N_0} M_i(h) \right| > C \{ \sqrt{v(J+x)} + B(J+x) \} \right] \leq C e^{-cx},$$

where

$$J = V \log(Ae) + V \log \left\{ \left(\frac{N_0 B^2}{v} \right) \vee 1 \right\}.$$

The constants $C, c > 0$ depend only on A, V .

PROOF. After division by B , it is enough to treat $B = 1$. Let ν_i be a regular conditional law of Z_i given $\mathcal{G}_{i-1}$, and put $Q_{N_0} = N_0^{-1} \sum_{i=1}^{N_0} \nu_i$. For $h, h' \in \mathcal{C}$,

$$\frac{1}{N_0} \sum_{i=1}^{N_0} \mathbb{E}\{[M_i(h) - M_i(h')]^2 \mid \mathcal{G}_{i-1}\} \leq 4 \|h - h'\|_{Q_{N_0}, 2}^2.$$

The entropy assumption therefore controls the predictable L^2 metrics of the martingale class. With $\sigma^2 = (v/N_0) \wedge 1$,

$$\int_0^\sigma \sqrt{V \log(Ae/u)} du \leq C \sigma J^{1/2}.$$

Substituting this entropy-integral bound, the quadratic-variation bound v , and the jump bound $|M_i(h)| \leq 2$ into the bounded martingale maximal inequalities of [van de Geer \(1995\)](#) and [Nishiyama \(2000\)](#) gives

$$\mathbb{P} \left[\sup_{h \in \mathcal{C}} \left| \sum_{i=1}^{N_0} M_i(h) \right| > C \{ \sqrt{v(J+x)} + J+x \} \right] \leq C e^{-cx}.$$

Rescaling restores (A.5). □

PROOF OF THEOREM 2.9. Write $A_s = (-\infty, s_1] \times \dots \times (-\infty, s_d]$. We first prove the bound on the localization cube K_{A_n} , where $A_n = A_n(b_n)$ is defined in (2.8), and then remove the localization.

Martingale–predictable decomposition. For $s \in \mathbb{R}^d$, define

$$D_i^{(L)}(s) = \mathbf{1}(X_i \leq s) - \mathbb{E}\{\mathbf{1}(X_i \leq s) \mid \xi_{i-L}\}.$$

Stationarity and the definition of F_L give

$$(A.6) \quad \mathbf{1}(X_i \leq s) - F(s) = D_i^{(L)}(s) + \{F_L(s \mid \xi_{i-L}) - F(s)\}.$$

Martingale component. Let

$$C_b = \{h_{s,t}(x) = \mathbf{1}(x \leq s) - \mathbf{1}(x \leq t) : s, t \in \mathbb{R}^d, |s - t|_\infty \leq b\}.$$

The fixed-dimensional lower-orthant class is VC, and the difference class $\mathcal{C}_b$ is VC-subgraph by the stability operations in [van der Vaart and Wellner \(1996, Lemma 2.6.20\)](#). Thus there are constants $A, V < \infty$, depending only on d , such that, for every probability measure Q and $0 < \epsilon < 1$,

$$(A.7) \quad N\{\epsilon \|H_b\|_{Q,2}, \mathcal{C}_b, L^2(Q)\} \leq (A/\epsilon)^V, \quad H_b \leq 2.$$

The usual rational-endpoint convention makes the class countable on each localization cube; the unrestricted supremum follows by monotone separability.

For $h_{s,t} \in \mathcal{C}_{b_n}$, the symmetric difference $A_s \triangle A_t$ is contained in the union of the coordinate slabs

$$\{s_j \wedge t_j < X_{i,j} \leq s_j \vee t_j\}, \quad j = 1, \dots, d.$$

Condition 2.6, shifted from time L to time i , therefore implies

$$(A.8) \quad \mathbb{E}\{h_{s,t}(X_i)^2 \mid \xi_{i-L}\} \leq \mathbb{P}(X_i \in A_s \triangle A_t \mid \xi_{i-L}) \leq Cb_n.$$

Fix a residue class $a = 1, \dots, L$. Write $i = a + kL$ and $\mathcal{G}_k^{(a)} = \sigma(\xi_{a+kL})$. Then

$$h(X_{a+kL}) - \mathbb{E}\{h(X_{a+kL}) \mid \mathcal{G}_{k-1}^{(a)}\}, \quad h \in \mathcal{C}_{b_n},$$

is a martingale-difference class, and (A.8) gives a predictable quadratic-variation bound $v = Cnb_n$. Applying Lemma A.2 with $B = 2$ gives

$$J \leq C \log(e/b_n).$$

Because $b_n \downarrow 0$, $\log(e/b_n) \lesssim \log n$ for all large n . The assumption $\log n = O(nb_n)$ absorbs the linear Bernstein term into the square-root term when $x = K \log n$. Summing the resulting probability bound over the fixed number L of residue classes yields

$$(A.9) \quad \sup_{\substack{s,t \in K_{A_n} \\ |s-t|_\infty \leq b_n}} \left| \sum_{i=1}^n \{D_i^{(L)}(s) - D_i^{(L)}(t)\} \right| = O_{\mathbb{P}}(\sqrt{nb_n \log n}).$$

Predictable component. For $s, t \in K_{A_n}$, connect the two corners by the coordinate path

$$v^{(0)} = s, \quad v^{(j)} = (t_1, \dots, t_j, s_{j+1}, \dots, s_d), \quad j = 1, \dots, d.$$

Condition 2.7 gives

$$(A.10) \quad \begin{aligned} & \sum_{i=1}^n \{F_L(t \mid \xi_{i-L}) - F_L(s \mid \xi_{i-L}) - [F(t) - F(s)]\} \\ &= \sum_{j=1}^d \int_{s_j}^{t_j} \sum_{i=1}^n \{q_{L,j}(z, v_{-j}^{(j)} \mid \xi_{i-L}) - \mathbb{E}q_{L,j}(z, v_{-j}^{(j)} \mid \xi_0)\} dz. \end{aligned}$$

Taking absolute values and using $|t_j - s_j| \leq b_n$,

$$\left| n^{-1/2} \sum_{i=1}^n \{F_L(t \mid \xi_{i-L}) - F_L(s \mid \xi_{i-L}) - [F(t) - F(s)]\} \right| \leq db_n R_n^\partial(A_n).$$

Condition 2.8 makes the right-hand side $O_{\mathbb{P}}(b_n L_n^\partial)$. Combining this bound with (A.6) and (A.9) gives the localized oscillation bound

$$(A.11) \quad \begin{aligned} & \sup_{\substack{s,t \in K_{A_n} \\ |s-t|_\infty \leq b_n}} \left| \frac{1}{\sqrt{n}} \sum_{i=1}^n [\mathbf{1}(X_i \leq s) - \mathbf{1}(X_i \leq t) - \{F(s) - F(t)\}] \right| \\ &= O_{\mathbb{P}}(\sqrt{b_n \log n}) + O_{\mathbb{P}}(b_n L_n^\partial). \end{aligned}$$

Tail localization. For $s \in \mathbb{R}^d$, let $\bar{s}$ be the coordinatewise clipping of s to K_{A_n} . Then $|\bar{s} - \bar{t}|_\infty \leq |s - t|_\infty$, and

$$A_s \triangle A_{\bar{s}} \subset \{x : |x|_\infty > A_n\}.$$

Consequently, uniformly in s ,

$$|S_n(1, s) - S_n(1, \bar{s})| \leq \sum_{i=1}^n \mathbf{1}(|X_i|_\infty > A_n) + n\mathbb{P}(|X_0|_\infty > A_n).$$

By Markov's inequality and (2.8),

$$\mathbb{P}(|X_0|_\infty > A_n) \leq \frac{\mathbb{E}|X_0|_\infty^p}{A_n^p} \leq C \sqrt{\frac{b_n \log n}{n}}.$$

The normalized tail contribution is therefore

$$n^{-1/2} \left\{ \sum_{i=1}^n \mathbf{1}(|X_i|_\infty > A_n) + n\mathbb{P}(|X_0|_\infty > A_n) \right\} = O_{\mathbb{P}}(\sqrt{b_n \log n}),$$

because its expectation is of that order. Combining the clipping inequality with (A.11) proves (2.9). $\square$

LEMMA A.3 (Coordinate slab consequence). *Under Theorem 2.9, if $b_n \downarrow 0$ and $\log n = O(nb_n)$, then*

$$\max_{1 \leq j \leq d} \sup_{a \in \mathbb{R}} \left| \sum_{i=1}^n [\mathbf{1}\{|X_{i,j} - a| \leq b_n\} - \mathbb{P}(|X_{0,j} - a| \leq b_n)] \right| = O_{\mathbb{P}}\{\sqrt{nb_n \log n} + \sqrt{n} b_n L_n^\partial\}.$$

PROOF. It is enough to control intervals $(a, a + h]$ with $0 < h \leq 2b_n$. Fix a coordinate j , and for $M < \infty$ define

$$s^{(M)} = (M, \dots, M, a, M, \dots, M), \quad t^{(M)} = (M, \dots, M, a + h, M, \dots, M),$$

where the displayed coordinate is the j th coordinate. Then

$$\mathbf{1}(X_i \leq t^{(M)}) - \mathbf{1}(X_i \leq s^{(M)}) = \mathbf{1}(a < X_{i,j} \leq a + h) \prod_{\ell \neq j} \mathbf{1}(X_{i,\ell} \leq M).$$

Theorem 2.9, applied with radius $2b_n$, controls the centered sum uniformly over a, h, M . Letting $M \rightarrow \infty$ along a countable sequence and using monotone convergence transfers this displayed bound to the coordinate interval $(a, a + h]$. Taking the supremum over rational a, h and using separability of monotone interval classes completes the proof. $\square$

A.3. Finite-grid coupling ingredients. In this subsection $S_n(f_s) = \sum_{i=1}^n f_s(X_i) = S_n(1, s)$ and

$$d_F(f_s, f_t) = \{\mathbb{E}[f_s(X_0) - f_t(X_0)]^2\}^{1/2}.$$

LEMMA A.4 (Tail-adapted lower-orthant grids). *Let $\delta \in (0, 1)$ and define*

$$(A.12) \quad A_\delta = \left\{ \frac{n(\mathbb{E}|X_0|_\infty^p)^2}{\delta^2 \log n} \right\}^{1/(2p)}.$$

There is a lattice $\Theta_\delta^{(d)} \subset [-A_\delta, A_\delta]^d$ such that each $s \in \mathbb{R}^d$ has a projection $\Pi_\delta s \in \Theta_\delta^{(d)}$ after coordinatewise clipping to $[-A_\delta, A_\delta]^d$, with $|\bar{s} - \Pi_\delta s|_\infty \leq C\delta^2$, and

$$(A.13) \quad |\Theta_\delta^{(d)}| \leq Cn^{d/(2p)} \delta^{-d(2+1/p)} (\log n)^{-d/(2p)}.$$

Moreover,

$$(A.14) \quad \mathbb{P}(|X_0|_\infty > A_\delta) \leq C \frac{\delta \sqrt{\log n}}{\sqrt{n}},$$

and if $|s - t|_\infty \leq C\delta^2$ inside the cube, then $d_F(f_s, f_t) \leq C\delta$.

PROOF. Markov's inequality and (A.12) give

$$\mathbb{P}(|X_0|_\infty > A_\delta) \leq \frac{\mathbb{E}|X_0|_\infty^p}{A_\delta^p} \leq C \frac{\delta \sqrt{\log n}}{\sqrt{n}}.$$

On the cube take a regular lattice with coordinate mesh $c\delta^2$. The number of grid points per coordinate is

$$O(A_\delta \delta^{-2}) = O\{n^{1/(2p)} \delta^{-(2+1/p)} (\log n)^{-1/(2p)}\},$$

which proves (A.13) after raising to the d th power. If $|s - t|_\infty \leq c\delta^2$, the symmetric difference $A_s \triangle A_t$ is contained in d coordinate slabs of width $c\delta^2$. The coordinate density bound in Condition 2.2 gives

$$d_F(f_s, f_t)^2 \leq \mathbb{P}(X_0 \in A_s \triangle A_t) \leq C_d c \delta^2.$$

Choosing c small and absorbing constants proves the metric claim. $\square$

LEMMA A.5 (Projection decomposition). *Let $G = \{s_1, \dots, s_M\}$ be a finite lower-orthant grid and write*

$$Y_i^G = (f_{s_1}(X_i), \dots, f_{s_M}(X_i)).$$

Assume

$$\mathcal{D}_2^\circ := \sum_{r=0}^{\infty} \sup_{s \in G} \|f_s(X_r) - f_s(X'_r)\|_2 < \infty$$

and

$$\mathcal{G}_2^\circ := \sum_{r=1}^{\infty} r \sup_{s \in G} \|f_s(X_r) - f_s(X'_r)\|_2 < \infty.$$

Define

$$(A.15) \quad D_i^G = \sum_{k=i}^{\infty} \mathcal{P}_i Y_k^G, \quad \mathcal{P}_i Z = \mathbb{E}(Z \mid \mathcal{F}_i) - \mathbb{E}(Z \mid \mathcal{F}_{i-1}).$$

Then $(D_i^G, \mathcal{F}_i)$ is an M -dimensional martingale-difference sequence and

$$(A.16) \quad \sum_{i=1}^n Y_i^G = \sum_{i=1}^n D_i^G + R_n^G,$$

where, coordinatewise,

$$(A.17) \quad \sup_{n \geq 1} \max_{1 \leq j \leq M} \|R_{n,j}^G\|_2 \leq 2\mathcal{G}_2^\circ.$$

PROOF. For $m \geq i$, the coupled variable obtained by replacing only ε_i gives

$$(A.18) \quad \sup_{s \in G} \|\mathcal{P}_i f_s(X_m)\|_2 \leq \sup_{s \in G} \|f_s(X_{m-i}) - f_s(X'_{m-i})\|_2.$$

By stationarity it is enough to prove this for $i = 0$. If $Y_m^*(s)$ is obtained from $Y_m(s) = f_s(X_m)$ by replacing ε_0 , then

$$\mathbb{E}\{Y_m^*(s) \mid \mathcal{F}_0\} = \mathbb{E}\{Y_m(s) \mid \mathcal{F}_{-1}\}.$$

Hence $\mathcal{P}_0 Y_m(s) = \mathbb{E}\{Y_m(s) - Y_m^*(s) \mid \mathcal{F}_0\}$, and conditional Jensen proves (A.18). Thus the series in (A.15) converges in L^2 , and its terms are $\mathcal{F}_i$ -measurable with conditional mean zero given $\mathcal{F}_{i-1}$.

For a fixed coordinate s_j , Fubini's theorem is justified by (A.18), and

$$\sum_{i=1}^n D_{i,j}^G = \sum_{k=1}^{\infty} \sum_{i=1}^{k \wedge n} \mathcal{P}_i f_{s_j}(X_k).$$

If $k \leq n$, the inner sum telescopes to

$$f_{s_j}(X_k) - \mathbb{E}\{f_{s_j}(X_k) \mid \mathcal{F}_0\};$$

if $k > n$, it telescopes to

$$\mathbb{E}\{f_{s_j}(X_k) \mid \mathcal{F}_n\} - \mathbb{E}\{f_{s_j}(X_k) \mid \mathcal{F}_0\}.$$

Consequently $R_{n,j}^G = Q_{0,j}^G - Q_{n,j}^G$, where

$$Q_{m,j}^G = \sum_{k=m+1}^{\infty} \mathbb{E}\{f_{s_j}(X_k) \mid \mathcal{F}_m\}.$$

Using the projection expansion of the conditional expectation,

$$\|Q_{m,j}^G\|_2 \leq \sum_{k \geq 1} \sum_{r \geq k} \sup_{s \in G} \|f_s(X_r) - f_s(X'_r)\|_2 = \mathcal{G}_2^{\circ},$$

uniformly in m, j , which proves (A.17). □

For a finite grid G , put

$$a_{u,G}(r) = \sup_{s \in G} \|f_s(X_r) - f_s(X'_r)\|_u, \quad \Theta_{u,G}(h) = \sum_{r=h}^{\infty} a_{u,G}(r).$$

LEMMA A.6 (Remote-innovation influence of D_i^G). Assume $\sum_{r=0}^{\infty} a_{u,G}(r) < \infty$. Let $D_h^{G,*(0)}$ be the coupled version of D_h^G obtained by replacing ε_0 . Then

$$(A.19) \quad \max_{s_j \in G} \|D_{h,j}^G - D_{h,j}^{G,*(0)}\|_u \leq 2\Theta_{u,G}(h).$$

PROOF. Fix $s_j \in G$. For $a \geq 0$, let $Y_{h+a}^{*(0)}(s_j)$ be the copy of $Y_{h+a}(s_j) = f_{s_j}(X_{h+a})$ after replacing ε_0 . The conditional future innovations after time h , given either $\mathcal{F}_h$ or $\mathcal{F}_{h-1}$, can be coupled before and after the replacement of ε_0 . Hence conditional Jensen gives

$$\|\mathcal{P}_h Y_{h+a}(s_j) - \{\mathcal{P}_h Y_{h+a}(s_j)\}^{*(0)}\|_u \leq 2 \|Y_{h+a}(s_j) - Y_{h+a}^{*(0)}(s_j)\|_u.$$

Minkowski's inequality and the definition of D_h^G imply

$$\|D_{h,j}^G - D_{h,j}^{G,*(0)}\|_u \leq 2 \sum_{a=0}^{\infty} \|Y_{h+a}(s_j) - Y_{h+a}^{*(0)}(s_j)\|_u \leq 2\Theta_{u,G}(h).$$

□

LEMMA A.7 (Covariance bound via projections). *Let (A_i) and (B_i) be centered, square-integrable, stationary causal sequences adapted to $(\mathcal{F}_i)$. Suppose $\mathbb{E}(A_0 \mid \mathcal{F}_{-\infty}) = \mathbb{E}(B_0 \mid \mathcal{F}_{-\infty}) = 0$, where $\mathcal{F}_{-\infty} = \cap_m \mathcal{F}_m$. Then, for every $h \geq 0$,*

$$(A.20) \quad |\text{Cov}(A_0, B_h)| \leq \sum_{r=0}^{\infty} \|\mathcal{P}_0 A_r\|_2 \|\mathcal{P}_0 B_{h+r}\|_2.$$

PROOF. By martingale convergence,

$$A_0 = \sum_{r=0}^{\infty} \mathcal{P}_{-r} A_0, \quad B_h = \sum_{r=0}^{\infty} \mathcal{P}_{-r} B_h$$

in L^2 . Orthogonality of projection increments gives

$$\text{Cov}(A_0, B_h) = \sum_{r=0}^{\infty} \mathbb{E}\{\mathcal{P}_{-r} A_0 \mathcal{P}_{-r} B_h\}.$$

Cauchy–Schwarz and stationarity yield (A.20). $\square$

LEMMA A.8 (Covariance identity and conditional-variance fluctuation). *Assume the hypotheses of Lemma A.5. Then*

$$(A.21) \quad \mathbb{E}\{D_0^G (D_0^G)^\top\} = \sum_{k \in \mathbb{Z}} \text{Cov}(Y_0^G, Y_k^G).$$

If, in addition, $\sum_{r=0}^{\infty} a_{4,G}(r) < \infty$ and

$$\sum_{h=0}^{\infty} \sum_{r=0}^{\infty} \Theta_{4,G}(r) \Theta_{4,G}(h+r) < \infty,$$

then, with

$$V_i^G = \text{Var}(D_i^G \mid \mathcal{F}_{i-1}), \quad \Omega_n^G = \sum_{i=1}^n V_i^G - n \mathbb{E}\{D_0^G (D_0^G)^\top\},$$

one has

$$(A.22) \quad \mathbb{E}|\Omega_n^G|_{\text{F}} \leq CM\sqrt{n}.$$

PROOF. For scalar coordinates $s, t \in G$, Lemma A.7 with $A_i = f_s(X_i)$ and $B_i = f_t(X_i)$, together with (A.18), gives

$$\sum_{h=0}^{\infty} |\text{Cov}\{f_s(X_0), f_t(X_h)\}| \leq \sum_{h=0}^{\infty} \sum_{r=0}^{\infty} a_{2,G}(r) a_{2,G}(h+r) \leq \left(\sum_{r=0}^{\infty} a_{2,G}(r) \right)^2.$$

Negative lags are handled by stationarity, so the long-run covariance is absolutely summable. Lemma A.5 gives

$$S_n(f_s) = \sum_{i=1}^n D_i(s) + R_n(s), \quad \sup_n \|R_n(s)\|_2 < \infty.$$

Martingale orthogonality implies

$$n^{-1} \mathbb{E} \left[\left(\sum_{i=1}^n D_i(s) \right) \left(\sum_{i=1}^n D_i(t) \right) \right] = \mathbb{E}\{D_0(s) D_0(t)\}.$$

The L^2 -bounded remainders disappear after division by n , while absolute summability gives

$$n^{-1}\mathbb{E}\{S_n(f_s)S_n(f_t)\} \rightarrow \sum_{k \in \mathbb{Z}} \text{Cov}\{f_s(X_0), f_t(X_k)\}.$$

This proves (A.21).

For the fluctuation bound, write

$$U_{i,j\ell} = \mathbb{E}(D_{i,j}^G D_{i,\ell}^G \mid \mathcal{F}_{i-1}) - \mathbb{E}(D_{i,j}^G D_{i,\ell}^G).$$

The (j, ℓ) entry of Ω_n^G is $\sum_{i=1}^n U_{i,j\ell}$. By Lemma A.6 with $u = 4$,

$$\sup_j \left\| D_{h,j}^G - D_{h,j}^{G,*(0)} \right\|_4 \leq C\Theta_{4,G}(h), \quad \sup_{h,j} \|D_{h,j}^G\|_4 \leq C\Theta_{4,G}(0).$$

After replacing ε_0 , the product $D_{h,j}^G D_{h,\ell}^G$ therefore changes in L^2 by at most

$$C\Theta_{4,G}(0)\Theta_{4,G}(h).$$

Conditional expectation and centering are L^2 -contractions, so

$$\sup_{j,\ell} \|\mathcal{P}_0 U_{h,j\ell}\|_2 \leq C\Theta_{4,G}(0)\Theta_{4,G}(h).$$

Applying Lemma A.7 to the stationary sequence $(U_{i,j\ell})$ and using the double summability assumption yields

$$\sup_{j,\ell} \mathbb{E} \left| \sum_{i=1}^n U_{i,j\ell} \right|^2 \leq Cn.$$

Therefore

$$\mathbb{E}|\Omega_n^G|_{\mathbb{F}} \leq M \sup_{j,\ell} \left\| \sum_{i=1}^n U_{i,j\ell} \right\|_2 \leq CM\sqrt{n}.$$

□

PROPOSITION A.9 (Finite-grid Gaussian coupling). *Let G be a finite lower-orthant grid with $M = |G|$. Under Conditions 2.2 and 2.4, one can construct a centered Gaussian vector $T_n^G = (T_n^G(s) : s \in G)$ with covariance*

$$(A.23) \quad \text{Cov}\{T_n^G(s), T_n^G(t)\} = n \sum_{k \in \mathbb{Z}} \text{Cov}\{f_s(X_0), f_t(X_k)\}$$

such that, for every deterministic $R_n \rightarrow \infty$,

$$(A.24) \quad \max_{s \in G} |S_n(f_s) - T_n^G(s)| = O_{\mathbb{P}} \left[\left\{ n^{1/3} M^{1/3} (\log M)^{5/6} + n^{1/4} M^{1/2} (\log M)^{1/2} \right\} R_n \right].$$

PROOF. Lemma A.1 and Condition 2.4 imply, for every $u \geq 2$,

$$\sup_{s \in G} \|f_s(X_r) - f_s(X_r')\|_u \leq C\rho_0^{r/u}$$

for some $\rho_0 \in (0, 1)$. Hence all projection sums in Lemmas A.5 and A.8 are finite. By Lemma A.5,

$$(A.25) \quad S_n^G := \sum_{i=1}^n Y_i^G = \sum_{i=1}^n D_i^G + R_n^G.$$

The martingale differences D_i^G are infinite projection sums, but the truncated sums $D_i^{G,L} = \sum_{k=i}^{i+L} \mathcal{P}_i Y_k^G$ converge to D_i^G in every fixed L^v . To justify applying the martingale coupling directly to D_i^G , apply the Yurinskii bound first to $(D_i^{G,L}, \mathcal{F}_i)_{i=1}^n$. The geometric projection tail implies

$$\max_{i \leq n} \max_{j \leq M} \|D_{i,j}^{G,L} - D_{i,j}^G\|_v \rightarrow 0 \quad (L \rightarrow \infty)$$

for every fixed $v \geq 2$. Hence the martingale sums converge in L^2 , the covariance matrices $n\mathbb{E}\{D_0^{G,L}(D_0^{G,L})^\top\}$ converge entrywise to Σ_G , the third-moment quantities in β_G converge by uniform L^4 -integrability, and the conditional-variance fluctuation matrices converge in L^1 by the same projection-tail bound applied to products. The endpoint projection remainders in Lemma A.5 also converge in L^2 . Passing to the limit along a coupling subsequence therefore gives the displayed bound for the infinite projection martingale.

We apply Cattaneo, Masini and Underwood (2025, Supplement, Lemma SA.8) to $(D_i^G, \mathcal{F}_i)_{i=1}^n$ with deterministic target covariance

$$\Sigma_G = n\mathbb{E}\{D_0^G(D_0^G)^\top\}.$$

Lemma A.8 identifies this matrix with (A.23). Let

$$V_i^G = \text{Var}(D_i^G | \mathcal{F}_{i-1}), \quad \Omega_n^G = \sum_{i=1}^n V_i^G - n\mathbb{E}\{D_0^G(D_0^G)^\top\}.$$

The second-order branch of the martingale Yurinskii lemma and the Gaussian maximal bound

$$\mathbb{E}|Z|_\infty \leq C\sqrt{\log(2M)}, \quad Z \sim N(0, I_M),$$

yield

$$\begin{aligned} \mathbb{P}\left(\max_{s \in G} |S_n(f_s) - T_n^G(s)| > \eta\right) &\leq C \frac{\{\beta_G \log(2M)\}^{1/3}}{\eta} \\ &+ C \left(\frac{\mathbb{E}|\Omega_n^G|_F \log(2M)}{\eta^2}\right)^{1/3} + \mathbb{P}(|R_n^G|_\infty > \eta/6), \end{aligned} \quad (\text{A.26})$$

where

$$\beta_G = \sum_{i=1}^n \mathbb{E}\left\{|D_i^G|_2^2 |D_i^G|_\infty + |(V_i^G)^{1/2} Z_i|_2^2 |(V_i^G)^{1/2} Z_i|_\infty\right\},$$

and $Z_i \stackrel{\text{iid}}{\sim} N(0, I_M)$, independently of the data.

The inherited geometric dependence gives

$$\max_{j \leq M} \|D_{i,j}^G\|_v \leq C \sum_{r=0}^{\infty} \rho_0^{r/v} \leq Cv, \quad v \geq 2.$$

Thus

$$\mathbb{E}|D_i^G|_\infty^2 \leq C(\log M)^2, \quad \max_{i,j} \mathbb{E}|D_{i,j}^G|^4 \leq C,$$

and

$$\sum_{i=1}^n \mathbb{E}\{|D_i^G|_2^2 |D_i^G|_\infty\} \leq CnM \log M.$$

For $G_i = (V_i^G)^{1/2} Z_i$, conditional Gaussian moment bounds give

$$\mathbb{E}(|G_i|_2^4 | \mathcal{F}_{i-1}) \leq 3\{\text{tr}(V_i^G)\}^2, \quad \mathbb{E}(|G_i|_\infty^2 | \mathcal{F}_{i-1}) \leq C \log(2M) \max_{j \leq M} (V_i^G)_{jj}.$$

Because $(V_i^G)_{jj} \leq \mathbb{E}\{(D_{i,j}^G)^2 | \mathcal{F}_{i-1}\}$, the preceding moment bound implies

$$\sum_{i=1}^n \mathbb{E}\{|G_i|_2^2 |G_i|_\infty\} \leq CnM(\log M)^{3/2}, \quad \beta_G \leq CnM(\log M)^{3/2}.$$

Lemma A.8 also gives $\mathbb{E}|\Omega_n^G|_F \leq CM\sqrt{n}$. Finally, the proof of Lemma A.5, with L^4 in place of L^2 , gives

$$|R_n^G|_\infty = O_{\mathbb{P}}(M^{1/4}R_n).$$

This is dominated by the rate in (A.24). Taking

$$\eta = CR_n\{n^{1/3}M^{1/3}(\log M)^{5/6} + n^{1/4}M^{1/2}(\log M)^{1/2}\}$$

in (A.26) proves the proposition. $\square$

LEMMA A.10 (Gaussian extension and canonical oscillation). *The finite-grid Gaussian vector in Proposition A.9 can be realized as the restriction of a centered Gaussian process $\{T_n(1, s) : s \in \mathbb{R}^d\}$ with covariance $K_n(s, t) = n\Gamma(s, t)$. Moreover, under Conditions 2.2 and 2.4, for every $0 < \tau < 1$ the process has a separable version satisfying*

$$(A.27) \quad \mathbb{E} \sup_{d_F(f_s, f_t) \leq \delta} |T_n(1, s) - T_n(1, t)| \leq C_\tau \sqrt{n} \delta^\tau \sqrt{\log(1/\delta)}$$

at the tail-adapted mesh scales used in Lemma A.4.

PROOF. The kernel Γ is positive semidefinite: for any finite coefficients a_j , Lemma A.8 gives

$$\sum_{j, \ell} a_j a_\ell \sum_{k \in \mathbb{Z}} \text{Cov}\{f_{s_j}(X_0), f_{s_\ell}(X_k)\} = \mathbb{E} \left(\sum_j a_j D_0(s_j) \right)^2 \geq 0.$$

Let $\mathcal{D}$ be a countable d_F -dense lower-orthant subclass that contains the finite grid G . Kolmogorov's extension theorem gives a centered Gaussian law on $\mathbb{R}^{\mathcal{D}}$ with covariance K_n . Since the marginal law on G agrees with T_n^G , we may enlarge the probability space and sample the remaining coordinates conditionally on T_n^G .

For the oscillation bound, set $\bar{Y}_r(s, t) = f_s(X_r) - f_t(X_r)$ and

$$a_r = \sup_u \|f_u(X_r) - f_u(X'_r)\|_2.$$

The projection bound gives both

$$\|\mathcal{P}_0 \bar{Y}_r(s, t)\|_2 \leq 2a_r, \quad \|\mathcal{P}_0 \bar{Y}_r(s, t)\|_2 \leq d_F(f_s, f_t).$$

Interpolating,

$$\|\mathcal{P}_0 \bar{Y}_r(s, t)\|_2 \leq C d_F(f_s, f_t)^\tau a_r^{1-\tau}.$$

Because a_r decays geometrically, Lemma A.7 implies

$$\sum_{h \in \mathbb{Z}} |\text{Cov}\{\bar{Y}_0(s, t), \bar{Y}_h(s, t)\}| \leq C_\tau d_F(f_s, f_t)^{2\tau}.$$

Thus the canonical metric of T_n is bounded by

$$d_{T_n}(s, t) = \{\mathbb{E}|T_n(1, s) - T_n(1, t)|^2\}^{1/2} \leq C_\tau \sqrt{n} d_F(f_s, f_t)^\tau.$$

The lower-orthant class and its difference class have polynomial L^2 -entropy by [van der Vaart and Wellner \(1996, Theorems 2.6.4 and 2.6.7, Lemma 2.6.20\)](#). Dudley's entropy integral ([van der Vaart and Wellner, 1996, Theorem 2.2.4](#)) gives

$$\mathbb{E} \sup_{d_F(f_s, f_t) \leq \delta} |T_n(1, s) - T_n(1, t)| \leq C \sqrt{n} \int_0^\delta u^{\tau-1} \sqrt{\log(1/u)} du,$$

which is bounded by (A.27). The estimate in (A.27) implies uniform continuity on localized classes and hence the separable extension to the d_F -closure. $\square$

A.4. End-point Gaussian approximation.

PROOF OF THEOREM 2.12. Fix $0 < \tau < 1$, to be chosen close to one at the end, and put $\delta = n^{-\lambda}$. Let $G_\delta = \Theta_\delta^{(d)}$ be the grid from Lemma A.4, and write $M_\delta = |G_\delta|$. Then

$$(A.28) \quad M_\delta \leq C n^{d/(2p)} \delta^{-d(2+1/p)} (\log n)^C.$$

Finite grid. By Proposition A.9, enlarged if necessary by Lemma A.10, we may construct a Gaussian process $T_n(1, \cdot)$ with covariance $K_n(s, t) = n\Gamma(s, t)$ such that

$$(A.29) \quad \begin{aligned} \max_{u \in G_\delta} |S_n(1, u) - T_n(1, u)| &= O_{\mathbb{P}}\{n^{1/3} M_\delta^{1/3} \log^C n + n^{1/4} M_\delta^{1/2} \log^C n\} \\ &= O_{\mathbb{P}}\{n^{A_{p,d}} \delta^{-B_{p,d}} \log^C n + n^{A'_{p,d}} \delta^{-B'_{p,d}} \log^C n\}, \end{aligned}$$

where

$$A_{p,d} = \frac{1}{3} + \frac{d}{6p}, \quad B_{p,d} = \frac{d(2p+1)}{3p},$$

and

$$A'_{p,d} = \frac{1}{4} + \frac{d}{4p}, \quad B'_{p,d} = \frac{d(2p+1)}{2p}.$$

Here we take the harmless diverging factor R_n in Proposition A.9 to be $\log n$, which is absorbed into the displayed $\log^C n$.

Empirical off-grid part. For $s \in \mathbb{R}^d$, let $\bar{s}$ be the coordinatewise clipping of s to $[-A_\delta, A_\delta]^d$, and let $\Pi_\delta s \in G_\delta$ be a nearest grid point to $\bar{s}$. Lemma A.4 gives

$$\sup_{s \in \mathbb{R}^d} |S_n(1, s) - S_n(1, \bar{s})| \leq \sum_{i=1}^n \mathbf{1}(|X_i|_\infty > A_\delta) + n\mathbb{P}(|X_0|_\infty > A_\delta),$$

whose expectation is $O\{\delta \sqrt{n \log n}\}$. Hence this clipping term is $O_{\mathbb{P}}\{\delta \sqrt{n \log n}\}$. On the cube, $|\bar{s} - \Pi_\delta s|_\infty \leq C\delta^2$. Theorem 2.9 with $b_n = C\delta^2$ applies because the final λ below is less than $1/2$, so $\log n = O(n\delta^2)$. Since Condition 2.8 gives $L_n^\partial = n^{o(1)}$,

$$(A.30) \quad \sup_{s \in \mathbb{R}^d} |S_n(1, \bar{s}) - S_n(1, \Pi_\delta s)| = O_{\mathbb{P}}\{\delta \sqrt{n \log n}\} + O_{\mathbb{P}}\{\sqrt{n} \delta^2 n^{o(1)}\}.$$

Gaussian off-grid part. The lattice step has d_F -diameter $O(\delta)$. The clipping step has

$$d_F(f_s, f_{\bar{s}}) \leq \mathbb{P}(|X_0|_\infty > A_\delta)^{1/2} \leq C\delta^{1/2} n^{-1/4} (\log n)^{1/4} \leq C\delta \log^C n$$

at the final mesh, because $\lambda < 1/2$. Lemma A.10 and Markov's inequality therefore give

$$(A.31) \quad \sup_{s \in \mathbb{R}^d} |T_n(1, s) - T_n(1, \Pi_\delta s)| = O_{\mathbb{P}}\{\sqrt{n} \delta^\tau \log^C n\}.$$

Combining (A.29), (A.30), and (A.31), we have

$$(A.32) \quad \sup_{s \in \mathbb{R}^d} |S_n(1, s) - T_n(1, s)| = O_{\mathbb{P}}[n^{A_{p,d}} \delta^{-B_{p,d}} \log^C n + n^{A'_{p,d}} \delta^{-B'_{p,d}} \log^C n + \sqrt{n} \delta^\tau \log^C n + \sqrt{n} \delta^2 n^{o(1)} + \delta \sqrt{n \log n}].$$

Balancing. Choose

$$(A.33) \quad \lambda = \lambda_\tau = \frac{1/2 - A_{p,d}}{\tau + B_{p,d}} = \frac{p - d}{2(2pd + 3p\tau + d)}.$$

The condition $p > d$ is exactly $\lambda_\tau > 0$, and $\lambda_\tau < 1/2$. The first finite-grid term and the Gaussian off-grid term in (A.32) are then both of order $n^{\gamma_{p,d,\tau}} \log^C n$, where

$$(A.34) \quad \gamma_{p,d,\tau} = A_{p,d} + B_{p,d} \lambda_\tau = \frac{1}{2} - \tau \lambda_\tau = \frac{\tau A_{p,d} + B_{p,d}/2}{\tau + B_{p,d}}.$$

The second finite-grid term is smaller, because

$$\gamma_{p,d,\tau} - \{A'_{p,d} + B'_{p,d} \lambda_\tau\} = \frac{\tau(p - d)}{4(2pd + 3p\tau + d)} > 0.$$

The empirical off-grid terms are also smaller: $1/2 - \lambda_\tau < 1/2 - \tau \lambda_\tau$ and $1/2 - 2\lambda_\tau < 1/2 - \tau \lambda_\tau$; the subpolynomial factor in (A.30) is absorbed into the final n^η slack. Finally,

$$\lim_{\tau \uparrow 1} \gamma_{p,d,\tau} = \frac{p + d(p + 1)}{3p + d(2p + 1)} = \gamma_0.$$

Given the $\eta > 0$ in the theorem, choose τ sufficiently close to one so that $\gamma_{p,d,\tau} < \gamma_0 + \eta/2$. The preceding bound then gives $O_{\mathbb{P}}(n^{\gamma_0 + \eta/2} \log^C n)$, and the fixed logarithmic power is absorbed into $n^{\eta/2}$. This proves (2.13). $\square$

APPENDIX B: INDEPENDENT-TAIL FINITE-WINDOW TRUNCATION

Let $\{\varepsilon_{i,j}^* : i \in \mathbb{Z}, j \leq i - 1\}$ be an array of i.i.d. copies of ε_0 , independent of the original innovations. Define

$$(B.1) \quad X_i^{[m]} = g(\dots, \varepsilon_{i,i-m-1}^*, \varepsilon_{i,i-m}^*, \varepsilon_{i-m+1}, \dots, \varepsilon_i).$$

The tail copies are indexed by i , so different observations use independent remote tails. This finite-window approximation follows the physical-dependence coupling method of Wu (2005b, 2008); it differs from conditional-expectation truncations because it preserves the marginal distribution exactly and hence keeps the empirical-indicator centering unchanged.

LEMMA B.1 (Basic properties). *For each $m \geq 1$, $(X_i^{[m]})_{i \in \mathbb{Z}}$ is stationary and m -dependent, and $X_i^{[m]} \stackrel{d}{=} X_i$. Let*

$$\delta_q(k) = \left\| |X_k - X_k^{*(0)}|_\infty \right\|_q$$

be the usual physical-dependence coefficient obtained by replacing ε_0 by an independent copy. Then

$$(B.2) \quad \left\| |X_i - X_i^{[m]}|_\infty \right\|_q \leq \Theta_q(m) := \sum_{r=m}^{\infty} \delta_q(r).$$

Consequently, under geometric decay $\delta_q(r) \leq C\rho^r$,

$$(B.3) \quad \left\| |X_i - X_i^{[m]}|_\infty \right\|_q \leq C\rho^m,$$

while under polynomial decay $\delta_q(r) \leq C(1+r)^{-\alpha}$ with $\alpha > 1$,

$$(B.4) \quad \left\| |X_i - X_i^{[m]}|_\infty \right\|_q \leq Cm^{-(\alpha-1)}.$$

PROOF. Stationarity and equality of marginals follow from the fact that the innovation sequence in (B.1) has the same product law as $(\dots, \varepsilon_{i-1}, \varepsilon_i)$. If $|i-j| \geq m$, the original innovation windows used by $X_i^{[m]}$ and $X_j^{[m]}$ are disjoint, and their remote tail arrays are independent; hence the process is m -dependent. The bound (B.2) follows by replacing the remote innovations one at a time and applying Minkowski's inequality. Replacing ε_{i-r} costs at most $\delta_q(r)$ by stationarity. The geometric and polynomial bounds follow by summing the corresponding tails. $\square$

Write from now on

$$(B.5) \quad X_i^{(m)} := X_i^{[m]}, \quad f_{m,s}(x) = \mathbf{1}(x \leq s) - F(s).$$

The centering is by F , because $X_i^{(m)} \stackrel{d}{=} X_i$.

For each fixed m define the enlarged filtration

$$(B.6) \quad \mathcal{G}_i^{(m)} = \sigma\{\varepsilon_r : r \leq i; \varepsilon_{\ell,r}^* : \ell \leq i, r \leq \ell - m\},$$

so that both X_i and $X_i^{(m)}$ are $\mathcal{G}_i^{(m)}$ -measurable. Let

$$(B.7) \quad \mathcal{P}_i^{(m)} Z = \mathbb{E}(Z | \mathcal{G}_i^{(m)}) - \mathbb{E}(Z | \mathcal{G}_{i-1}^{(m)}).$$

The additional tail variables are independent of the original process, so projection identities for the original process are unchanged after enlarging the filtration.

APPENDIX C: TERMINAL COUPLING FOR FINITE-WINDOW BLOCKS

The next lemma gives a terminal coupling for truncated blocks. It uses two uniform bounds: all dependence on m is bounded by m^2 , and the probability parameter enters as $R^{3/2}$. The latter gives a direct treatment of the conditional-variance fluctuation in the martingale Yurinskii coupling.

LEMMA C.1 (Terminal coupling for m -dependent truncations). *Let $Y_i = X_i^{(m)}$ be the independent-tail truncation (B.1). Suppose $N \geq 2$ and $1 \leq m \leq N^C$ for a fixed constant $C < \infty$. Assume that d is fixed, $\mathbb{E}|Y_0|_\infty^p < \infty$ for some $p > d$, and the coordinate marginal densities of Y_0 are uniformly bounded. Let*

$$\Gamma_m(s, t) = \sum_{|h| \leq m} \text{Cov}\{f_{m,s}(Y_0), f_{m,t}(Y_h)\}.$$

For every $R \geq 1$ and every $\eta > 0$, there exists a centered Gaussian process $G_N^{(m)}$ with covariance

$$\text{Cov}\{G_N^{(m)}(s), G_N^{(m)}(t)\} = N\Gamma_m(s, t)$$

such that

$$(C.1) \quad \mathbb{P} \left(\sup_{s \in \mathbb{R}^d} \left| \sum_{i=1}^N f_{m,s}(Y_i) - G_N^{(m)}(s) \right| > CN^{\gamma_0 + \eta} m^2 R^{3/2} \log^C N \right) \leq \frac{C}{R}.$$

The constants may depend on d, p and the primitive moment and density bounds, but not on N, m, R .

PROOF. We prove the result in four steps. Constants may change from line to line. The argument gives a uniform high-probability block coupling; sharp dependence on m is not needed for the sequential rate balance.

Step 1: finite-grid martingale reduction. Let $G = \{s_1, \dots, s_M\} \subset \mathbb{R}^d$ be finite and write

$$Y_i^G = (f_{m,s_1}(Y_i), \dots, f_{m,s_M}(Y_i)).$$

Use the filtration $\mathcal{G}_i^{(m)}$ and projections $\mathcal{P}_i^{(m)}$. Since Y_k depends on the original innovations only through the window $\{\varepsilon_{k-m+1}, \dots, \varepsilon_k\}$ and on a tail array indexed by k , it is independent of the time- i innovation increment of the enlarged filtration whenever $k \geq i + m + 1$. Hence the projection series is finite. Define

$$(C.2) \quad D_i^G = \sum_{r=0}^m \mathcal{P}_i^{(m)} Y_{i+r}^G.$$

Then $(D_i^G, \mathcal{G}_i^{(m)})$ is an M -dimensional martingale-difference sequence. Interchanging the two finite sums in $\sum_i D_i^G$ and telescoping the projection increments gives

$$(C.3) \quad \sum_{i=1}^N Y_i^G = \sum_{i=1}^N D_i^G + R_N^G,$$

where R_N^G contains only projection layers that cross the left and right endpoints of $\{1, \dots, N\}$. More explicitly, after interchanging the two finite sums, all interior terms telescope, while the unmatched left endpoint terms have indices $k \leq 0$ and the unmatched right endpoint terms have indices $k > N$, with at most m projection layers and at most $m+1$ future observations in each layer. Since $|Y_{i,j}^G| \leq 1$ and conditional expectation is an L^v -contraction,

$$(C.4) \quad |D_i^G|_\infty \leq Cm, \quad \max_{j \leq M} \|R_{N,j}^G\|_v \leq C_v m^2, \quad v \geq 2.$$

Consequently, by the union bound and Markov's inequality with $v = 4$,

$$(C.5) \quad |R_N^G|_\infty = O_P(M^{1/4} m^2 R^{1/4})$$

with probability loss at most C/R . This remainder is smaller than the coupling threshold used below.

The projection covariance identity is exact in the finite-dependent case:

$$(C.6) \quad \mathbb{E}\{D_0^G (D_0^G)^\top\} = \sum_{h \in \mathbb{Z}} \text{Cov}(Y_0^G, Y_h^G).$$

Indeed, fix m and G , and let $N' \rightarrow \infty$ be an auxiliary sample size used only for this identity. The covariance sum is finite because (Y_i) is m -dependent. The projection representation above gives

$$(N')^{-1} \text{Var} \left(\sum_{i=1}^{N'} D_i^G \right) = \mathbb{E}\{D_0^G (D_0^G)^\top\}.$$

On the other hand, the covariance of $\sum_{i=1}^{N'} Y_i^G$, divided by N' , converges to the finite sum on the right side of (C.6). The endpoint remainder in (C.3) has L^2 -norm $O(m^2)$ coordinatewise for fixed m , and therefore disappears after division by N' . Letting $N' \rightarrow \infty$ proves (C.6). Thus the deterministic target covariance for the martingale coupling is exactly the long-run covariance matrix Γ_m restricted to G .

Step 2: Yurinskii inputs on the grid. Let

$$V_i^G = \text{Var}(D_i^G | \mathcal{G}_{i-1}^{(m)}), \quad \Omega_N^G = \sum_{i=1}^N V_i^G - N \mathbb{E}\{D_0^G (D_0^G)^\top\}.$$

For each pair (j, k) , the process $\{(V_i^G)_{jk} - \mathbb{E}(V_i^G)_{jk}\}$ is finite dependent with dependence range at most a constant multiple of m . This is because D_i^G is a sum of $m+1$ projections involving only $Y_i, \dots, Y_{i+m}$, and each of these variables is m -dependent under the independent-tail construction. Moreover,

$$|(V_i^G)_{jk}| \leq \mathbb{E}(|D_{i,j}^G D_{i,k}^G| | \mathcal{G}_{i-1}^{(m)}) \leq C m^2$$

by (C.4). Therefore

$$\mathbb{E} \left| \sum_{i=1}^N \{(V_i^G)_{jk} - \mathbb{E}(V_i^G)_{jk}\} \right|^2 \leq C N m^5.$$

Summing over j, k gives

$$(C.7) \quad \mathbb{E} |\Omega_N^G|_F \leq C M \sqrt{N} m^{5/2}.$$

Next consider the third-moment quantity in the martingale Yurinskii inequality. Since $|D_i^G|_\infty \leq C m$,

$$\sum_{i=1}^N \mathbb{E}\{|D_i^G|_2^2 |D_i^G|_\infty\} \leq C N M m^3.$$

For the conditional Gaussian term, let $Z_i \sim N(0, I_M)$ independent of the data and set $G_i = (V_i^G)^{1/2} Z_i$. Conditionally on $\mathcal{G}_{i-1}^{(m)}$, the coordinate variances of G_i are bounded by $C m^2$, while $\text{tr}(V_i^G) \leq C M m^2$. Standard Gaussian maximal inequalities imply

$$\mathbb{E}\{|G_i|_2^2 |G_i|_\infty | \mathcal{G}_{i-1}^{(m)}\} \leq C M m^3 \sqrt{\log(2M)}.$$

Thus the full third-moment input satisfies

$$(C.8) \quad \beta_G \leq C N M m^3 \log^C(2M).$$

We use the vector-valued martingale Yurinskii coupling of Cattaneo, Masini and Underwood (Cattaneo, Masini and Underwood, 2025, Supplement, Lemma SA.8, and the simplification leading to Corollary 2.2). Applied to the martingale array $(D_i^G, \mathcal{G}_i^{(m)})_{i=1}^N$, with deterministic covariance $N \Gamma_m|_{G \times G}$, the theorem bounds the grid coupling by a third-moment input and a conditional-variance fluctuation input. These two quantities are β_G in (C.8) and Ω_N^G in (C.7). Combining the resulting probability bound with these two displays, and then adding the projection remainder bound (C.5), yields a centered Gaussian vector $G_N^{(m)}(G)$ such that

$$(C.9) \quad \mathbb{P} \left(\max_{s \in G} \left| \sum_{i=1}^N f_{m,s}(Y_i) - G_N^{(m)}(s) \right| > C R^{3/2} \{N^{1/3} M^{1/3} m + N^{1/4} M^{1/2} m^{5/4} + M^{1/4} m^2\} \log^C(2M) \right) \leq \frac{C}{R}.$$

The power $R^{3/2}$ is chosen to dominate the third-moment branch, the conditional-variance branch, and the projection remainder. Since $m \geq 1$, the right side of (C.9) is bounded by

$$(C.10) \quad CR^{3/2}m^2\{N^{1/3}M^{1/3} + N^{1/4}M^{1/2} + M^{1/4}\}\log^C(2M).$$

Step 3: tail-adapted grid and off-grid oscillations. Let

$$D_{p,d} = 3p + d(2p + 1), \quad \lambda = \frac{p-d}{2D_{p,d}}, \quad \delta = N^{-\lambda}.$$

This is the mesh exponent that balances the leading grid-coupling term with the Gaussian off-grid term. Let A_δ be the tail radius

$$A_\delta = \left\{ \frac{N(\mathbb{E}|X_0|_\infty^p)^2}{\delta^2 \log N} \right\}^{1/(2p)}.$$

Since $Y_0 \stackrel{d}{=} X_0$,

$$(C.11) \quad \mathbb{P}(|Y_0|_\infty > A_\delta) \leq C \frac{\delta \sqrt{\log N}}{\sqrt{N}}.$$

On $[-A_\delta, A_\delta]^d$ choose a lattice G_δ with coordinate mesh $c\delta^2$. Its cardinality satisfies

$$(C.12) \quad M := |G_\delta| \leq CN^{d/(2p)}\delta^{-d(2+1/p)}\log^C N.$$

With the preceding λ ,

$$N^{d/(2p)}\delta^{-d(2+1/p)} = N^{d(p+2)/D_{p,d}}.$$

Therefore

$$\frac{1}{3} + \frac{d(p+2)}{3D_{p,d}} = \frac{p+d(p+1)}{D_{p,d}} = \gamma_0,$$

while

$$\gamma_0 - \left\{ \frac{1}{4} + \frac{d(p+2)}{2D_{p,d}} \right\} = \frac{p-d}{4D_{p,d}} > 0, \quad \gamma_0 - \frac{d(p+2)}{4D_{p,d}} > 0.$$

Consequently,

$$(C.13) \quad N^{1/3}M^{1/3} + N^{1/4}M^{1/2} + M^{1/4} \leq CN^{\gamma_0+\eta/4}\log^C N.$$

The strict inequality $p > d$ is used in the second display and is the point where the moment threshold enters the grid coupling.

For the empirical off-grid term, split $\{1, \dots, N\}$ into residue classes modulo $m+1$. Within each residue class the variables Y_i are independent. The lower-orthant difference class

$$\mathcal{C}_\delta = \{x \mapsto \mathbf{1}(x \leq s) - \mathbf{1}(x \leq t) : |s - t|_\infty \leq C\delta^2\}$$

is VC and has variance bounded by $C\delta^2$ on the localized cube, because $A_s \triangle A_t$ is contained in a union of d coordinate slabs of width $C\delta^2$ and the coordinate marginal densities are uniformly bounded. Apply Lemma A.2 to each independent residue class, with variance proxy $C(N/m+1)\delta^2$, envelope 2, and VC entropy constants depending only on d . The one-residue bound is $O_P\{\sqrt{(N/m+1)}\delta\log^C N + \log^C N\}$; summing over the $m+1$ residue classes gives

$$(C.14) \quad \sup_{\substack{s, t \in [-A_\delta, A_\delta]^d \\ |s-t|_\infty \leq C\delta^2}} \left| \sum_{i=1}^N \{f_{m,s}(Y_i) - f_{m,t}(Y_i)\} \right| = O_P\{R\sqrt{Nm}\delta\log^C N + Rm\log^C N\}$$

with probability loss C/R . By (C.11), the contribution outside the cube $[-A_\delta, A_\delta]^d$ has expectation

$$O(\delta\sqrt{N\log N}),$$

and is absorbed by (C.14).

For the Gaussian off-grid term, the covariance kernel satisfies

$$(C.15) \quad \mathbb{E}|G_N^{(m)}(s) - G_N^{(m)}(t)|^2 \leq CNm \mathbb{P}(Y_0 \in A_s \triangle A_t).$$

Indeed, the variance is

$$N \sum_{|h| \leq m} \text{Cov}\{g_{s,t}(Y_0), g_{s,t}(Y_h)\}, \quad g_{s,t} = \mathbf{1}(\cdot \leq s) - \mathbf{1}(\cdot \leq t) - \{F(s) - F(t)\}.$$

The centered function $g_{s,t}$ need not vanish outside $A_s \triangle A_t$. Instead,

$$\text{Var}\{g_{s,t}(Y_0)\} = \text{Var}\{\mathbf{1}(Y_0 \leq s) - \mathbf{1}(Y_0 \leq t)\} \leq \mathbb{P}(Y_0 \in A_s \triangle A_t).$$

By Cauchy–Schwarz and stationarity,

$$|\text{Cov}\{g_{s,t}(Y_0), g_{s,t}(Y_h)\}| \leq \mathbb{P}(Y_0 \in A_s \triangle A_t), \quad |h| \leq m.$$

Summing the $O(m)$ nonzero lags gives (C.15). The VC entropy of lower orthants and Dudley’s integral therefore imply

$$(C.16) \quad \mathbb{E} \sup_{\substack{s,t \in [-A_\delta, A_\delta]^d \\ |s-t|_\infty \leq C\delta^2}} |G_N^{(m)}(s) - G_N^{(m)}(t)| \leq C\sqrt{Nm} \delta \log^C N.$$

It remains to control the Gaussian clipping outside the localization cube. For $s \in \mathbb{R}^d$, let $\bar{s}$ be the coordinatewise clipping of s to $[-A_\delta, A_\delta]^d$. Since $A_s \triangle A_{\bar{s}} \subset \{|Y_0|_\infty > A_\delta\}$, the same argument that gives (C.15) yields

$$\mathbb{E}|G_N^{(m)}(s) - G_N^{(m)}(\bar{s})|^2 \leq CNm \mathbb{P}(|Y_0|_\infty > A_\delta).$$

The class $\{\mathbf{1}(\cdot \leq s) - \mathbf{1}(\cdot \leq \bar{s}) : s \in \mathbb{R}^d\}$ is a VC-subgraph difference class, and (C.11) gives

$$\mathbb{E} \sup_{s \in \mathbb{R}^d} |G_N^{(m)}(s) - G_N^{(m)}(\bar{s})| \leq C\sqrt{Nm \mathbb{P}(|Y_0|_\infty > A_\delta)} \log^C N \leq Cm^{1/2} N^{1/4} \delta^{1/2} \log^C N.$$

Since $\delta = N^{-\lambda}$ with $\lambda > 0$, the power $1/4 - \lambda/2$ is strictly smaller than γ_0 . Markov’s inequality therefore shows that the clipping term is also dominated by the uniform threshold below.

Markov’s inequality adds a factor R and probability loss C/R . Because $R \geq 1$ and $m \geq 1$, all empirical and Gaussian off-grid and clipping terms are dominated by the bound $CR^{3/2}m^2 N^{\gamma_0+\eta} \log^C N$.

Step 4: extension to the continuum. The covariance kernel $N\Gamma_m$ is positive semidefinite, so the finite-grid Gaussian vector can be realized as the restriction of a separable Gaussian process on a countable dense subclass of lower orthants. The preceding Gaussian continuity bound extends it to all $s \in \mathbb{R}^d$ under the standard VC separability convention. Combining the grid bound (C.10), the rate calculation (C.13), and the empirical/Gaussian off-grid bounds proves (C.1). $\square$

APPENDIX D: GEOMETRIC PROOF

Choose block parameters

$$(D.1) \quad a = 2\gamma_{\text{seq}}, \quad \ell_n = \lfloor n^a \rfloor, \quad m_n = \lfloor C_m \log n \rfloor, \quad h_n = \lfloor C_h \log n \rfloor,$$

with $C_h > C_m + 2$. Let

$$(D.2) \quad \zeta_n = \rho^{qm_n/(q+1)} = n^{-A+o(1)},$$

where C_m is chosen so that

$$(D.3) \quad 1 - \gamma_{\text{seq}} < A < 1.$$

The upper bound $A < 1$ guarantees the side condition $\log n = O(n\zeta_n)$ in Lemma A.3. Split time into alternating big and small blocks

$$I_j = \{a_j + 1, \dots, a_j + \ell_n\}, \quad J_j = \{a_j + \ell_n + 1, \dots, a_j + \ell_n + h_n\},$$

where $a_j = (j-1)(\ell_n + h_n)$, and let $N_n^{\text{blk}} = \max\{j : a_j + \ell_n \leq n\} \asymp n/\ell_n$. Define

$$(D.4) \quad \mathcal{B}_j(s) = \sum_{i \in I_j} f_{m_n, s}(X_i^{(m_n)}).$$

Because $h_n \geq m_n$, the process-valued blocks $\mathcal{B}_1, \dots, \mathcal{B}_{N_n^{\text{blk}}}$ are independent.

LEMMA D.1 (Geometric uniform truncation).

$$(D.5) \quad \sup_{0 \leq k \leq n} \sup_{s \in \mathbb{R}^d} \left| \sum_{i=1}^k \{f_s(X_i) - f_{m_n, s}(X_i^{(m_n)})\} \right| = o_P(n^{\gamma_{\text{seq}}}).$$

PROOF. Set $D_i(s) = \mathbf{1}(X_i \leq s) - \mathbf{1}(X_i^{(m_n)} \leq s)$. Then

$$f_s(X_i) - f_{m_n, s}(X_i^{(m_n)}) = D_i(s) - \mathbb{E}D_i(s).$$

If $D_i(s) \neq 0$, then either $|X_i - X_i^{(m_n)}|_\infty > \zeta_n$ or a coordinate boundary is crossed within distance ζ_n :

$$|D_i(s)| \leq \mathbf{1}\{|X_i - X_i^{(m_n)}|_\infty > \zeta_n\} + \sum_{j=1}^d \mathbf{1}\{|X_{i,j} - s_j| \leq \zeta_n\}.$$

By Lemma B.1 and Markov's inequality,

$$\sum_{i=1}^n \mathbf{1}\{|X_i - X_i^{(m_n)}|_\infty > \zeta_n\} = O_P(n\rho^{qm_n}\zeta_n^{-q}) = O_P(n\zeta_n).$$

The coordinate slab terms are controlled by Lemma A.3 and the bounded coordinate densities:

$$\sup_a \sum_{i=1}^n \mathbf{1}\{|X_{i,j} - a| \leq \zeta_n\} = O_P\{n\zeta_n + \sqrt{n\zeta_n \log n} + \sqrt{n}\zeta_n L_n^\partial\}.$$

The same bound also controls $n \sup_s |\mathbb{E}D_i(s)|$, because the expectation is bounded by the probability of the same crossing event and the marginal coordinate densities are bounded. Since $\zeta_n = n^{-A+o(1)}$ and $A > 1 - \gamma_{\text{seq}}$, all terms are $o_P(n^{\gamma_{\text{seq}}})$. $\square$

LEMMA D.2 (Projection residual diameter, geometric case). *For $m = m_n$, define*

$$D_s^{0,m} = \sum_{k=0}^{\infty} \mathcal{P}_0^{(m)} f_s(X_k), \quad D_{m,s} = \sum_{k=0}^{\infty} \mathcal{P}_0^{(m)} f_{m,s}(X_k^{(m)}),$$

and $H_m = \sup_s \|D_s^{0,m} - D_{m,s}\|_2$. Then for every fixed $C_0 < \infty$,

$$(D.6) \quad H_{m_n} = o(n^{\gamma_{\text{seq}} - 1/2} \log^{-C_0} n).$$

PROOF. Let $h_{m,s}(k) = f_s(X_k) - f_{m,s}(X_k^{(m)})$. By the same crossing argument,

$$\sup_s \|h_{m,s}(k)\|_2 \leq C\rho^{cm}.$$

For $0 \leq k \leq m$ this gives $\sup_s \|\mathcal{P}_0^{(m)} h_{m,s}(k)\|_2 \leq C\rho^{cm}$. For $k > m$, $X_k^{(m)}$ is independent of the time-zero projection increment of the enlarged filtration, hence

$$\sup_s \|\mathcal{P}_0^{(m)} h_{m,s}(k)\|_2 = \sup_s \|\mathcal{P}_0^{(m)} f_s(X_k)\|_2 \leq C\rho^{ck}.$$

Therefore $H_m \leq C m \rho^{cm} + C \sum_{k>m} \rho^{ck}$. With $m = m_n = C_m \log n$ and C_m sufficiently large, (D.6) follows. $\square$

LEMMA D.3 (Geometric small-block and within-block errors). *With the geometric block parameters,*

$$(D.7) \quad \sup_{0 \leq k \leq n} \sup_s \left| \sum_{i \leq k, i \in \cup_j J_j} f_{m_n,s}(X_i^{(m_n)}) \right| = o_P(n^{\gamma_{\text{seq}}}),$$

and

$$(D.8) \quad \max_{j \leq N_n^{\text{blk}} + 1} \sup_{0 \leq r \leq \ell_n} \sup_s \left| \sum_{\substack{i \in I_j, i \leq a_j + r \\ i \leq n}} f_{m_n,s}(X_i^{(m_n)}) \right| = O_P(\sqrt{\ell_n} \log^C n).$$

PROOF. The small-block sum is bounded deterministically by

$$N_n^{\text{blk}} h_n + h_n \lesssim n^{1-a} \log n = o(n^{\gamma_{\text{seq}}}),$$

because $a = 2\gamma_{\text{seq}}$ and $\gamma_{\text{seq}} > 1/3$. For (D.8), split indices into residues modulo $m_n + 1$. Within each residue the variables are independent. The extra index $N_n^{\text{blk}} + 1$ covers the possible terminal incomplete big block; the additional restriction $i \leq n$ only reduces the support. The class of functions

$$\{(i, x) \mapsto \mathbf{1}(i \in I_j, i \leq a_j + r, i \leq n) f_{m_n,s}(x) : j \leq N_n^{\text{blk}} + 1, 0 \leq r \leq \ell_n, s \in \mathbb{R}^d\}$$

is VC-subgraph on the product of the deterministic index and the observation space. In one residue class there are $N \asymp n/m_n$ independent observations, and every function is supported on at most $C\ell_n/m_n + C$ indices. Hence its normalized variance is $O(\ell_n/n)$. The localized VC maximal inequality for independent triangular arrays gives expectation $C\sqrt{\ell_n/m_n} \log^C n$ in one residue. Summing over $m_n + 1$ residues gives $C\sqrt{\ell_n m_n} \log^C n = C\sqrt{\ell_n} \log^C n$, and Markov's inequality proves the result. $\square$

LEMMA D.4 (Geometric block coupling). *For every $\eta > 0$, one can construct independent Gaussian processes $W_1, \dots, W_{N_n^{\text{blk}}}$ with covariance*

$$\text{Cov}\{W_j(s), W_j(t)\} = \ell_n \Gamma_{m_n}(s, t)$$

such that

$$(D.9) \quad \max_{b \leq N_n^{\text{blk}}} \sup_s \left| \sum_{j=1}^b \{\mathcal{B}_j(s) - W_j(s)\} \right| = O_P(n^{\gamma_{\text{seq}} + \eta} \log^C n).$$

PROOF. Apply Lemma C.1 with $N = \ell_n$, $m = m_n = O(\log \ell_n)$, and $R = N_n^{\text{blk}} n^{\eta_0}$. A union bound gives, with probability tending to one,

$$\max_{j \leq N_n^{\text{blk}}} \sup_s |\mathcal{B}_j(s) - W_j(s)| \leq C \ell_n^{\gamma_0 + \eta_0} m_n^2 (N_n^{\text{blk}} n^{\eta_0})^{3/2} \log^C n.$$

The accumulated error is at most N_n^{blk} times this display. Since $N_n^{\text{blk}} \asymp n/\ell_n$ and $\ell_n = n^{a+o(1)}$, the accumulated leading exponent is

$$\frac{5}{2}(1-a) + a\gamma_0 = \frac{5}{2} - a \left(\frac{5}{2} - \gamma_0 \right).$$

By (2.12), this equals γ_{seq} . Taking η_0 sufficiently small and absorbing $m_n^2 \asymp \log^2 n$ into $\log^C n$ proves the claim. $\square$

APPENDIX E: GAUSSIAN STITCHING

All projection sums in this section use $\mathcal{P}_i^{(m)}$ from (B.7). Let $\mathcal{H}_0$ be the closed linear span in L^2 of all $D_s^{0,m}$ and $D_{m,s}$. The projection covariance identity gives

$$(E.1) \quad \Gamma(s, t) = \left\langle D_s^{0,m}, D_t^{0,m} \right\rangle_{L^2}, \quad \Gamma_m(s, t) = \langle D_{m,s}, D_{m,t} \rangle_{L^2}.$$

Let $\mathbb{W}$ be an isonormal Gaussian process over $L^2[0, 1] \otimes \mathcal{H}_0$ and define

$$(E.2) \quad T_n(u, s) = \sqrt{n} \mathbb{W}(\mathbf{1}_{[0,u]} \otimes D_s^{0,m}), \quad T_n^{(m)}(u, s) = \sqrt{n} \mathbb{W}(\mathbf{1}_{[0,u]} \otimes D_{m,s}).$$

Then T_n has covariance (2.6), and $T_n^{(m)}$ has covariance $n(u \wedge v) \Gamma_m(s, t)$.

LEMMA E.1 (Gaussian residual comparison). *If $H_m = \sup_s \|D_s^{0,m} - D_{m,s}\|_2$ and $m \leq n^C$, then*

$$(E.3) \quad \mathbb{E} \sup_{u \in [0,1]} \sup_s |T_n(u, s) - T_n^{(m)}(u, s)| \leq C \sqrt{n} H_m \log^C n.$$

In particular, if $\sqrt{n} H_m \log^C n = o(n^r)$, then the residual is $o_P(n^r)$.

PROOF. Put $h_{m,s} = D_s^{0,m} - D_{m,s}$. The residual process is

$$R_n(u, s) = \sqrt{n} \mathbb{W}(\mathbf{1}_{[0,u]} \otimes h_{m,s}).$$

Its canonical metric satisfies

$$d_R\{(u, s), (v, t)\} \leq \sqrt{n} H_m |u - v|^{1/2} + \sqrt{n} \|h_{m,s} - h_{m,t}\|_2.$$

The full-sample projection metric bound gives, for some fixed $\tau \in (0, 1)$,

$$\|D_s^{0,m} - D_t^{0,m}\|_2 \leq C d_F(f_s, f_t)^\tau.$$

For the truncated m -dependent process the projection sum has at most $m + 1$ nonzero lags, and the same interpolation argument gives

$$\|D_{m,s} - D_{m,t}\|_2 \leq C m^C d_F(f_s, f_t)^\tau.$$

Therefore

$$(E.4) \quad \|h_{m,s} - h_{m,t}\|_2 \leq C m^C d_F(f_s, f_t)^\tau.$$

At the same time the class $\{h_{m,s} : s \in \mathbb{R}^d\}$ has L^2 diameter at most $2H_m$. To make the entropy step explicit, cover the lower-orthant class in d_F at radius

$$u = \left\{ \frac{\epsilon}{C \sqrt{n} m^C} \right\}^{1/\tau}.$$

The VC entropy bound for lower orthants gives a polynomial-size d_F -cover, and (E.4) turns it into an L^2 -cover of $\sqrt{n} \{h_{m,s}\}$ at radius ϵ . Intersecting this with the trivial diameter bound $2\sqrt{n}H_m$, and using $m \leq n^C$, yields, for $0 < \epsilon \leq C\sqrt{n}H_m$,

$$\log N\{\epsilon, \sqrt{n}\{h_{m,s} : s \in \mathbb{R}^d\}, L^2\} \leq C \log \left(\frac{C n^C}{\epsilon} \vee e \right).$$

The time index $u \mapsto \mathbf{1}_{[0,u]}$ contributes the same polynomial entropy under the metric $|u - v|^{1/2}$. Dudley's entropy integral for the product class yields

$$\mathbb{E} \sup_{u,s} |R_n(u, s)| \leq C \int_0^{C\sqrt{n}H_m} \sqrt{\log \left(\frac{C n^C}{\epsilon} \vee e \right)} d\epsilon \leq C \sqrt{n} H_m \log^C n.$$

This proves (E.3); Markov's inequality gives the probability statement. $\square$

LEMMA E.2 (Gaussian stitching, general form). *Let $W_n^{\text{blk}}(b, s) = \sum_{j=1}^b W_j(s)$, let $b(u)$ be the number of complete big blocks before time $\lfloor nu \rfloor$, and assume $m_n \leq n^C$ for a fixed $C < \infty$. Then there is a centered Gaussian process T_n with covariance (2.6) such that*

$$(E.5) \quad \begin{aligned} & \sup_{u \in [0,1]} \sup_s |W_n^{\text{blk}}(b(u), s) - T_n(u, s)| \\ &= O_{\mathbb{P}} \left(\sqrt{\ell_n m_n} \log^C n + \sqrt{n m_n h_n / \ell_n} \log^C n \right) + O_{\mathbb{P}}(\sqrt{n} H_{m_n} \log^C n). \end{aligned}$$

PROOF. At the reduced-clock endpoint $u_b = b\ell_n/n$,

$$\text{Cov}\{T_n^{(m)}(u_a, s), T_n^{(m)}(u_b, t)\} = (a \wedge b) \ell_n \Gamma_m(s, t),$$

which matches the covariance of $W_n^{\text{blk}}(a, s)$ and $W_n^{\text{blk}}(b, t)$. Hence we may realize

$$W_n^{\text{blk}}(b, s) = T_n^{(m)}(b\ell_n/n, s), \quad b = 0, 1, \dots, N_n^{\text{blk}}.$$

The mismatch between u and $b(u)\ell_n/n$ is at most

$$\Delta_n \leq C \left(\frac{\ell_n}{n} + \frac{N_n^{\text{blk}} h_n}{n} \right) \leq C \left(\frac{\ell_n}{n} + \frac{h_n}{\ell_n} \right).$$

For fixed s , the variance of a time increment of $T_n^{(m)}$ is bounded by $Cn|u - v|m_n$, because the long-run covariance of an m_n -dependent bounded process is $O(m_n)$. We record the uniform version of this modulus. Let

$$\mathcal{I}_{\Delta_n} = \{(a, b] : 0 \leq a \leq b \leq 1, b - a \leq \Delta_n\}.$$

The increment process can be written as

$$\sqrt{n} \mathbb{W}(\mathbf{1}_I \otimes D_{m,s}), \quad I \in \mathcal{I}_{\Delta_n}, \quad s \in \mathbb{R}^d.$$

Its canonical diameter is at most $C\sqrt{nm_n\Delta_n}$, since $\|D_{m,s}\|_2^2 = \Gamma_m(s, s) \leq Cm_n$. Moreover, for some fixed $\tau \in (0, 1)$,

$$\|D_{m,s} - D_{m,t}\|_2 \leq Cm_n^C d_F(f_s, f_t)^\tau,$$

by the same projection interpolation used in Lemma E.1. The interval class $\mathcal{I}_{\Delta_n}$ has polynomial $L^2[0, 1]$ -entropy, and the lower-orthant class has polynomial d_F -entropy. Therefore, for $0 < \epsilon \leq C\sqrt{nm_n\Delta_n}$,

$$\log N\left(\epsilon, \{\sqrt{n} \mathbf{1}_I \otimes D_{m,s} : I \in \mathcal{I}_{\Delta_n}, s \in \mathbb{R}^d\}, L^2[0, 1] \otimes L^2\right) \leq C \log\left(\frac{Cn^C}{\epsilon} \vee e\right).$$

Dudley's entropy integral gives

$$\mathbb{E} \sup_{|u-v| \leq \Delta_n} \sup_s |T_n^{(m)}(u, s) - T_n^{(m)}(v, s)| \leq C\sqrt{nm_n\Delta_n} \log^C n.$$

Markov's inequality then yields

$$\sup_{|u-v| \leq \Delta_n} \sup_s |T_n^{(m)}(u, s) - T_n^{(m)}(v, s)| = O_{\mathbb{P}}(\sqrt{nm_n\Delta_n} \log^C n).$$

Since

$$\sqrt{nm_n\Delta_n} \leq C \left(\sqrt{\ell_n m_n} + \sqrt{nm_n h_n / \ell_n} \right),$$

this gives the first two terms in (E.5). The last term follows from Lemma E.1. $\square$

PROOF OF THEOREM 2.13. Let $S_n^{\text{big}}(b, s) = \sum_{j=1}^b \mathcal{B}_j(s)$ and

$$\tilde{S}_n(k, s) = \sum_{i=1}^k f_{m_n, s}(X_i^{(m_n)}), \quad 0 \leq k \leq n.$$

Decompose

$$S_n(u, s) - T_n(u, s) = R_n^{\text{red}}(u, s) + E_n^{\text{blk}}(u, s) + G_n^{\text{stitch}}(u, s),$$

where

$$R_n^{\text{red}}(u, s) = S_n(u, s) - S_n^{\text{big}}(b(u), s),$$

$$E_n^{\text{blk}}(u, s) = S_n^{\text{big}}(b(u), s) - W_n^{\text{blk}}(b(u), s),$$

$$G_n^{\text{stitch}}(u, s) = W_n^{\text{blk}}(b(u), s) - T_n(u, s).$$

We first bound the reduction term. Put $k = \lfloor nu \rfloor$ and $b = b(u)$. By the definition of $b(u)$, the first k observations contain exactly the complete big blocks $I_1, \dots, I_b$, together with small-block observations and possibly one incomplete big block I_{b+1} . Therefore

$$\begin{aligned} \tilde{S}_n(k, s) - S_n^{\text{big}}(b, s) &= \sum_{\substack{i \leq k \\ i \in \bigcup_{j \geq 1} J_j}} f_{m_n, s}(X_i^{(m_n)}) \\ &\quad + \sum_{\substack{i \in I_{b+1} \\ i \leq k}} f_{m_n, s}(X_i^{(m_n)}), \end{aligned}$$

where the second sum is interpreted as zero if the block I_{b+1} has not started by time k . Since $b+1 \leq N_n^{\text{blk}}+1$, Lemma D.3 controls both displayed terms uniformly over $k \leq n$ and $s \in \mathbb{R}^d$. Lemma D.1 controls $\sup_{k,s} |S_n(k/n, s) - \tilde{S}_n(k, s)|$. Hence

$$\sup_{u,s} |R_n^{\text{red}}(u, s)| = O_P(\sqrt{\ell_n} \log^C n) + o_P(n^{\gamma_{\text{seq}}}) = O_P(n^{\gamma_{\text{seq}}} \log^C n).$$

Lemma D.4 controls E_n^{blk} . For G_n^{stitch} , Lemmas D.2 and E.2, with $m_n, h_n = O(\log n)$ and $\ell_n = n^{2\gamma_{\text{seq}}+o(1)}$, give $O_{\mathbb{P}}(n^{\gamma_{\text{seq}}} \log^C n)$. Combining the three bounds gives

$$\sup_{u,s} |S_n(u, s) - T_n(u, s)| = O_P(n^{\gamma_{\text{seq}}+\eta/2} \log^C n)$$

after applying Lemma D.4 with $\eta/2$ in place of η . Absorbing the fixed logarithmic power into $n^{\eta/2}$ proves (2.14). $\square$

APPENDIX F: PROOF OF THE POLYNOMIAL THEOREM

Throughout this section assume Conditions 2.2, 2.5, 2.6, 2.7, and 2.8, together with (2.17). The key difference from the geometric case is that the truncation length must grow polynomially, so the m -dependent block coupling, within-block oscillation, small-block deletion, and Gaussian stitching all pay powers of m_n . The next lemma provides the exponent choice used below.

LEMMA F.1 (Polynomial block-exponent feasibility). *Assume $\beta > \beta_{\text{crit}}(p, d)$, with β and β_{crit} defined in (2.15) and (2.16). Then there exist $0 < b < a < 1$ and $r < 1/2$ such that*

$$(F.1) \quad r > \frac{5}{2} - a \left(\frac{5}{2} - \gamma_0 \right) + 2b, \quad \text{block coupling accumulation,}$$

$$(F.2) \quad r > \frac{a+b}{2}, \quad \text{within-block and main clock modulus,}$$

$$(F.3) \quad r > 1 - a + b, \quad \text{small-block deletion,}$$

$$(F.4) \quad r > \frac{1+2b-a}{2}, \quad \text{deleted-clock Gaussian modulus,}$$

$$(F.5) \quad r > 1 - b\beta, \quad \text{uniform truncation,}$$

$$(F.6) \quad r > \frac{1}{2} - b\lambda_H, \quad \text{Gaussian covariance residual,}$$

$$(F.7) \quad b\beta < 1. \quad \text{local-oscillation side condition.}$$

PROOF. Put $C_\gamma = 5/2 - \gamma_0$ and set

$$a^\circ = \frac{8}{9 - 2\gamma_0}, \quad u^\circ = 1 - a^\circ = \frac{1 - 2\gamma_0}{9 - 2\gamma_0}.$$

Since $0 \leq \gamma_0 < 1/2$, we have $0 < u^\circ \leq 1/9$ and $a^\circ \in [8/9, 1)$. A direct calculation gives

$$\frac{a^\circ C_\gamma - 2}{2} = u^\circ, \quad 1 - a^\circ = u^\circ, \quad \beta_{\text{crit}}(p, d) = \frac{1}{2u^\circ}.$$

The strict inequality $\beta > \beta_{\text{crit}}(p, d)$ implies $1/(2\beta) < u^\circ$. Choose b such that

$$\frac{1}{2\beta} < b < \min \left\{ u^\circ, \frac{1}{\beta} \right\},$$

and put $a = a^\circ$. Then $0 < b < a < 1$ and $b\beta < 1$, which proves (F.7). We now check the six exponents that must lie below $1/2$. For (F.1),

$$\frac{5}{2} - aC_\gamma + 2b = \frac{1}{2} - 2(u^\circ - b) < \frac{1}{2}.$$

For (F.2),

$$\frac{a+b}{2} = \frac{1-u^\circ+b}{2} < \frac{1}{2},$$

because $b < u^\circ$. For (F.3) and (F.4),

$$1 - a + b = u^\circ + b < 2u^\circ \leq \frac{2}{9} < \frac{1}{2}, \quad \frac{1+2b-a}{2} = \frac{u^\circ+2b}{2} < \frac{3u^\circ}{2} \leq \frac{1}{6} < \frac{1}{2}.$$

For (F.5), the lower bound $b > 1/(2\beta)$ gives

$$1 - b\beta < \frac{1}{2}.$$

Finally, since $\beta_{\text{crit}}(p, d) \geq 9/2$, the condition $\beta > \beta_{\text{crit}}(p, d)$ implies $\lambda_H = \beta/2 - 1 > 0$; hence

$$\frac{1}{2} - b\lambda_H < \frac{1}{2}.$$

Let M be the maximum of the six quantities on the right-hand sides of (F.1)–(F.6). The preceding displays show that $M < 1/2$. Choosing any $r \in (M, 1/2)$ proves (F.1)–(F.6), and the lemma follows. $\square$

For the rest of this section, fix a, b, r satisfying (F.1)–(F.7); this r is one admissible value of r_{poly} in Theorem 2.14. Let β and λ_H be as in (2.15), and choose the polynomial block parameters

$$(F.8) \quad \ell_n = \lfloor n^a \rfloor, \quad m_n = \lfloor n^b \rfloor, \quad h_n = \lfloor n^b \rfloor, \quad 0 < b < a < 1.$$

The small-block length is chosen of the same order as the truncation length. As in the geometric proof, set

$$\begin{aligned} I_j &= \{a_j + 1, \dots, a_j + \ell_n\}, \\ J_j &= \{a_j + \ell_n + 1, \dots, a_j + \ell_n + h_n\}, \\ a_j &= (j-1)(\ell_n + h_n), \end{aligned}$$

and $N_n^{\text{blk}} = \max\{j : a_j + \ell_n \leq n\} \asymp n/\ell_n$. Since $h_n = m_n$, the big blocks based on the m_n -dependent truncation are independent.

With these parameters, define

$$(F.9) \quad \vartheta_m := \left\| |X_i - X_i^{(m)}|_\infty \right\|_q \leq C m^{-(\alpha-1)}, \quad \zeta_m := \vartheta_m^{q/(q+1)} \leq C m^{-\beta}.$$

For $m = m_n$, this gives $\zeta_{m_n} \leq C n^{-b\beta}$.

LEMMA F.2 (Polynomial uniform truncation). *Under (F.5) and (F.7),*

$$(F.10) \quad \sup_{0 \leq k \leq n} \sup_s \left| \sum_{i=1}^k \{f_s(X_i) - f_{m_n, s}(X_i^{(m_n)})\} \right| = o_P(n^r).$$

PROOF. Set $D_i(s) = \mathbf{1}(X_i \leq s) - \mathbf{1}(X_i^{(m_n)} \leq s)$. If $D_i(s) \neq 0$, then either the truncation error exceeds ζ_{m_n} , or one coordinate of X_i lies within ζ_{m_n} of the boundary s_j . Markov's inequality gives

$$\sum_{i=1}^n \mathbf{1}\{|X_i - X_i^{(m_n)}|_\infty > \zeta_{m_n}\} = O_P(n\vartheta_{m_n}^q \zeta_{m_n}^{-q}) = O_P(n\zeta_{m_n}).$$

The coordinate slab terms are controlled by Lemma A.3; the side condition $\log n = O(n\zeta_{m_n})$ follows from $b\beta < 1$, and Condition 2.8 gives $L_n^\partial = n^{o(1)}$. Hence

$$\sup_{k,s} \left| \sum_{i=1}^k \{f_s(X_i) - f_{m_n,s}(X_i^{(m_n)})\} \right| = O_P\{n\zeta_{m_n} + \sqrt{n\zeta_{m_n} \log n} + \sqrt{n\zeta_{m_n}} L_n^\partial\}.$$

Also,

$$\sup_s |\mathbb{E}D_i(s)| \leq \mathbb{P}(|X_i - X_i^{(m_n)}|_\infty > \zeta_{m_n}) + \sum_{j=1}^d \sup_a \mathbb{P}(|X_{i,j} - a| \leq \zeta_{m_n}),$$

so the bounded coordinate densities and the preceding Markov bound give the same order for $n \sup_s |\mathbb{E}D_i(s)|$. The leading exponent is $1 - b\beta$; the term $\sqrt{n\zeta_{m_n}} L_n^\partial$ has exponent $1/2 - b\beta + o(1)$, which is smaller. Hence (F.5) proves the lemma. $\square$

LEMMA F.3 (Polynomial projection residual). *Under (F.6),*

$$(F.11) \quad \sqrt{n} H_{m_n} \log^C n = o(n^r).$$

PROOF. For $0 \leq k \leq m$, the crossing bound gives

$$\sup_s \left\| \mathcal{P}_0^{(m)} \{f_s(X_k) - f_{m,s}(X_k^{(m)})\} \right\|_2 \leq C\zeta_m^{1/2} \leq Cm^{-\beta/2}.$$

For $k > m$, $X_k^{(m)}$ is independent of the time-zero projection increment of the enlarged filtration. The indicator-inheritance bound gives

$$\sup_s \left\| \mathcal{P}_0^{(m)} f_s(X_k) \right\|_2 \leq Ck^{-\alpha q/[2(q+1)]}.$$

Thus

$$H_m \leq Cm^{1-\beta/2} + C \sum_{k>m} k^{-\alpha q/[2(q+1)]}.$$

Let $\chi = \alpha q/[2(q+1)]$. Since

$$\chi = \frac{\beta}{2} + \frac{q}{2(q+1)}$$

and $\lambda_H = \beta/2 - 1 > 0$, we have $\chi > 1$ and

$$\sum_{k>m} k^{-\chi} \leq Cm^{1-\chi} = Cm^{-\lambda_H - q/[2(q+1)]} \leq Cm^{-\lambda_H}.$$

The first term is also $Cm^{-\lambda_H}$. Therefore $H_{m_n} \leq Cn^{-b\lambda_H}$. Condition (F.6) gives $1/2 - b\lambda_H < r$, so (F.11) follows. $\square$

LEMMA F.4 (Polynomial block reduction bounds). *The following bounds hold:*

$$(F.12) \quad \sup_{0 \leq k \leq n} \sup_s \left| \sum_{i \leq k, i \in \cup_j J_j} f_{m_n, s}(X_i^{(m_n)}) \right| = O(n^{1-a+b+o(1)}),$$

$$(F.13) \quad \max_{j \leq N_n^{\text{blk}} + 1} \sup_{0 \leq u \leq \ell_n} \sup_s \left| \sum_{\substack{i \in I_j, i \leq a_j + u \\ i \leq n}} f_{m_n, s}(X_i^{(m_n)}) \right| = O_{\mathbb{P}}(n^{(a+b)/2} \log^C n).$$

Consequently both are $o_P(n^r)$ under (F.2) and (F.3).

PROOF. The small-block bound is deterministic: $N_n^{\text{blk}} h_n \asymp n^{1-a+b}$. For the within-block term, repeat the proof of (D.8), now with $m_n + 1 \asymp n^b$ residue classes. The extra index $N_n^{\text{blk}} + 1$ covers the possible terminal incomplete big block, and the restriction $i \leq n$ only reduces the support. In each residue class the localized VC maximal bound is $O_P(\sqrt{\ell_n/m_n} \log^C n)$. Summing over the $m_n + 1$ residues gives

$$O_P(\sqrt{\ell_n m_n} \log^C n) = O_P(n^{(a+b)/2} \log^C n).$$

□

LEMMA F.5 (Polynomial block coupling). *For every $\eta > 0$ one can construct independent Gaussian processes $W_1, \dots, W_{N_n^{\text{blk}}}$ with covariance $\ell_n \Gamma_{m_n}$ such that*

$$(F.14) \quad \max_{b \leq N_n^{\text{blk}}} \sup_s \left| \sum_{j=1}^b \{\mathcal{B}_j(s) - W_j(s)\} \right| = O_{\mathbb{P}}(n^{r+\eta} \log^C n).$$

PROOF. Apply Lemma C.1 to each big block with $N = \ell_n$, $m = m_n$, and $R = N_n^{\text{blk}} n^{\eta_0}$. With probability tending to one,

$$\max_{j \leq N_n^{\text{blk}}} \sup_s |\mathcal{B}_j(s) - W_j(s)| \leq C \ell_n^{\gamma_0 + \eta_0} m_n^2 (N_n^{\text{blk}} n^{\eta_0})^{3/2} \log^C n.$$

Accumulating over at most N_n^{blk} blocks gives exponent

$$\frac{5}{2} - a \left(\frac{5}{2} - \gamma_0 \right) + 2b$$

up to arbitrarily small losses. This is smaller than r by (F.1); hence (F.14) follows. □

LEMMA F.6 (Polynomial Gaussian stitching). *With the polynomial block parameters,*

$$(F.15) \quad \sup_{u \in [0,1]} \sup_s |W_n^{\text{blk}}(b(u), s) - T_n(u, s)| = O_{\mathbb{P}}(n^{r+\eta} \log^C n)$$

for every $\eta > 0$.

PROOF. Use Lemma E.2. The first stitching term has exponent $(a+b)/2$, which is smaller than r by (F.2). The deleted-clock term has exponent $(1+2b-a)/2$, which is smaller than r by (F.4). The residual term is $o_P(n^r)$ by Lemma F.3. This proves the claim. □

PROOF OF THEOREM 2.14.

Let

$$\tilde{S}_n(k, s) = \sum_{i=1}^k f_{m_n, s}(X_i^{(m_n)}), \quad 0 \leq k \leq n,$$

and write the approximation error as

$$S_n(u, s) - T_n(u, s) = R_n^{\text{red}}(u, s) + E_n^{\text{blk}}(u, s) + G_n^{\text{stitch}}(u, s).$$

Here $S_n^{\text{big}}(b, s) = \sum_{j=1}^b \mathcal{B}_j(s)$,

$$R_n^{\text{red}}(u, s) = S_n(u, s) - S_n^{\text{big}}(b(u), s),$$

$$E_n^{\text{blk}}(u, s) = S_n^{\text{big}}(b(u), s) - W_n^{\text{blk}}(b(u), s),$$

$$G_n^{\text{stitch}}(u, s) = W_n^{\text{blk}}(b(u), s) - T_n(u, s).$$

To control R_n^{red} , put $k = \lfloor nu \rfloor$ and $b = b(u)$. The same deterministic block decomposition as in the geometric proof gives

$$\begin{aligned} \tilde{S}_n(k, s) - S_n^{\text{big}}(b, s) &= \sum_{\substack{i \leq k \\ i \in \bigcup_{j \geq 1} J_j}} f_{m_n, s}(X_i^{(m_n)}) \\ &\quad + \sum_{\substack{i \in I_{b+1} \\ i \leq k}} f_{m_n, s}(X_i^{(m_n)}), \end{aligned}$$

with the second sum interpreted as zero if I_{b+1} has not started. Since $b + 1 \leq N_n^{\text{blk}} + 1$, Lemma F.4 controls the small-block and terminal incomplete-block terms uniformly over $k \leq n$ and $s \in \mathbb{R}^d$. Lemma F.2 controls $\sup_{k, s} |S_n(k/n, s) - \tilde{S}_n(k, s)|$. Thus

$$\sup_{u, s} |R_n^{\text{red}}(u, s)| = o_P(n^r).$$

The accumulated block coupling term is controlled by Lemma F.5. The Gaussian stitching term G_n^{stitch} is controlled by Lemma F.6. Combining the three bounds, applying Lemmas F.5 and F.6 with $\eta/2$, and absorbing the fixed logarithmic power into $n^{\eta/2}$, prove (2.18) with $r_{\text{poly}} = r$. $\square$

APPENDIX G: PROOFS OF CONSEQUENCES AND MULTIPLIER CALIBRATION

PROOF OF COROLLARY 3.1. Under the geometric assumptions, Theorem 2.13 gives

$$(G.1) \quad \sup_{u, s} |S_n(u, s) - T_n(u, s)| = O_{\mathbb{P}}(n^{\gamma_{\text{seq}} + \eta}).$$

Under the polynomial assumptions, Theorem 2.14 gives

$$(G.2) \quad \sup_{u, s} |S_n(u, s) - T_n(u, s)| = O_{\mathbb{P}}(n^{r_{\text{poly}} + \eta}).$$

Equations (G.1) and (G.2) can be written as

$$(G.3) \quad \sup_{u, s} |S_n(u, s) - T_n(u, s)| = O_{\mathbb{P}}(n^{\theta + \eta}), \quad \theta < 1/2, \quad \eta < 1/2 - \theta.$$

By the triangle inequality and the bridge definitions in (3.3),

$$(G.4) \quad |\mathbb{D}_n(u, s) - \mathbb{B}_n(u, s)| \leq n^{-1/2} |S_n(u, s) - T_n(u, s)| + un^{-1/2} |S_n(1, s) - T_n(1, s)|.$$

Taking the supremum over $u \in [0, 1]$ and $s \in \mathbb{R}^d$ in (G.4), and then applying (G.3), proves (3.4).

For the covariance, expand

$$\begin{aligned} & \text{Cov}\{\mathbb{B}_n(u, s), \mathbb{B}_n(v, t)\} \\ (G.5) \quad &= n^{-1} \text{Cov}\{T_n(u, s) - uT_n(1, s), T_n(v, t) - vT_n(1, t)\}. \end{aligned}$$

Using $\text{Cov}\{T_n(a, s), T_n(b, t)\} = n(a \wedge b)\Gamma(s, t)$ in (G.5) gives

$$\{u \wedge v - uv - uv + uv\}\Gamma(s, t) = (u \wedge v - uv)\Gamma(s, t),$$

which is (3.5). $\square$

PROOF OF COROLLARY 3.2. Let $a_n = n^{\theta-1/2+\eta}$. By Corollary 3.1, $a_n = o(1)$ and

$$|CP_n - G_n^{\text{cp}}| = O_{\mathbb{P}}(a_n).$$

Choose a deterministic sequence $\delta_n \downarrow 0$ such that $\mathbb{P}(|CP_n - G_n^{\text{cp}}| > \delta_n) \rightarrow 0$. Then

$$\{G_n^{\text{cp}} \leq c_{n,1-\alpha} - \delta_n\} \subset \{CP_n \leq c_{n,1-\alpha}\} \subset \{G_n^{\text{cp}} \leq c_{n,1-\alpha} + \delta_n\}$$

up to an event whose probability tends to zero. The assumed continuity of G_n^{cp} at $c_{n,1-\alpha}$ and the definition of the critical value give (3.8). $\square$

PROOF OF LEMMA 3.5. Let $\mathcal{F}_i = \sigma(\varepsilon_r : r \leq i)$ and $\mathcal{P}_i Z = \mathbb{E}(Z \mid \mathcal{F}_i) - \mathbb{E}(Z \mid \mathcal{F}_{i-1})$. Write $g_{x,y} = f_x - f_y$ and

$$d_F(x, y) = \|g_{x,y}(X_0)\|_2 \leq |F(x) - F(y)|^{1/2}.$$

For $k \geq 0$, let

$$a_2(k) = \sup_{z \in \mathbb{R}} \|f_z(X_k) - f_z(X_k^{*(0)})\|_2.$$

The projection bound used in Lemma A.5 gives

$$\|\mathcal{P}_0 g_{x,y}(X_k)\|_2 \leq C \min\{d_F(x, y), a_2(k)\}.$$

Under geometric physical dependence, Lemma A.1 gives $a_2(k) \leq C\rho^k$. Hence, for every $\tau \in (0, 1)$,

$$\sum_{k=0}^{\infty} \min\{d_F(x, y), a_2(k)\} \leq C_{\tau} d_F(x, y)^{\tau}.$$

Under the polynomial assumptions,

$$a_2(k) \leq C(1+k)^{-\chi}, \quad \chi = \frac{\alpha q}{2(q+1)}.$$

The polynomial condition in Theorem 2.14 implies $\beta > \beta_{\text{crit}}(p, d) \geq 9/2$. Since

$$\chi = \frac{\beta}{2} + \frac{q}{2(q+1)},$$

we have $\chi > 1$. Consequently

$$\sum_{k=0}^{\infty} \min\{d_F(x, y), a_2(k)\} \leq C d_F(x, y)^{1-1/\chi}.$$

Thus, in either case, there is $\tau_0 > 0$ such that

$$\sum_{k=0}^{\infty} \|\mathcal{P}_{0g_{x,y}}(X_k)\|_2 \leq C d_F(x, y)^{\tau_0}.$$

The projection covariance bound then gives

$$\sum_{h \in \mathbb{Z}} |\text{Cov}\{g_{x,y}(X_0), g_{x,y}(X_h)\}| \leq C d_F(x, y)^{2\tau_0} \leq C |F(x) - F(y)|^{\tau_0}.$$

Also $\sup_x \sum_{h \in \mathbb{Z}} |\text{Cov}\{f_x(X_0), f_x(X_h)\}| < \infty$. Since

$$\text{Cov}\{Z_n(u, x), Z_n(v, y)\} = (u \wedge v) \Gamma(x, y),$$

the preceding displays imply

$$\|Z_n(u, x) - Z_n(u, y)\|_2 \leq C |F(x) - F(y)|^{\tau_0/2}, \quad \|Z_n(u, x) - Z_n(v, x)\|_2 \leq C |u - v|^{1/2}.$$

The triangle inequality proves the displayed canonical-metric bound with $\kappa = \tau_0/2$. $\square$

PROOF OF LEMMA 3.6. The class of time intervals $[0, u]$ and the class of half-lines $(-\infty, x]$ have polynomial entropy. Under the displayed canonical metric bound, Dudley's entropy integral gives

$$\mathbb{E} \sup_{u \in [0, 1], x \in \mathbb{R}} |Z_n(u, x)| \leq C,$$

after the usual separability reduction to rational u and x . Hence $\sup_{u, x} |Z_n(u, x)| = O_{\mathbb{P}}(1)$.

For the local modulus, the density of F is bounded on the quantile neighborhood, so for $|x - q_\tau| \leq M/\sqrt{n}$,

$$|F(x) - F(q_\tau)| \leq C_M n^{-1/2}.$$

The canonical diameter of the local class

$$\{(u, x), (u, q_\tau) : u \in [0, 1], \tau \in \mathcal{T}, |x - q_\tau| \leq M/\sqrt{n}\}$$

is therefore $O(n^{-\kappa/2})$. The same polynomial entropy bound, now localized to this shrinking diameter, yields

$$\mathbb{E} \sup_{u, \tau} \sup_{|x - q_\tau| \leq M/\sqrt{n}} |Z_n(u, x) - Z_n(u, q_\tau)| \leq C n^{-\kappa/2} \log^C n = o(1).$$

Markov's inequality proves (3.9). $\square$

PROOF OF COROLLARY 3.7. Choose $\eta > 0$ so small that $\theta + \eta < 1/2$. The corresponding sequential Gaussian approximation theorem gives

$$(G.6) \quad \sup_{u \in [0, 1]} \sup_{x \in \mathbb{R}} |n^{-1/2} S_n(u, x) - Z_n(u, x)| = O_{\mathbb{P}}(n^{\theta-1/2+\eta}) = o_{\mathbb{P}}(1).$$

Let $G_n(u, x) = n^{-1/2} S_n(u, x)$. Combining (G.6) with Lemmas 3.5 and 3.6 gives

$$(G.7) \quad \sup_{u \in [0, 1]} \sup_{x \in \mathbb{R}} |G_n(u, x)| = O_{\mathbb{P}}(1).$$

We first localize the sample quantiles. Uniformly over $u \in [u_0, 1]$ and $\tau \in \mathcal{T}$,

$$(G.8) \quad |\hat{q}_{u, \tau} - q_\tau| = O_{\mathbb{P}}(n^{-1/2}).$$

Indeed, for M large and $x = q_\tau + M/\sqrt{n}$, Taylor's theorem and the density lower bound give

$$(G.9) \quad F(x) - F(q_\tau) \geq c_f M/\sqrt{n} - CM^2/n.$$

The stochastic term

$$k_u^{-1} \sum_{i \leq k_u} \{\mathbf{1}(X_i \leq x) - F(x)\} = (k_u/\sqrt{n})^{-1} G_n(u, x)$$

is $O_{\mathbb{P}}(n^{-1/2})$ uniformly by (G.7) and $k_u \geq u_0 n/2$. For sufficiently large M , (G.9) dominates this stochastic term with probability tending to one. The lower side, with $x = q_\tau - M/\sqrt{n}$, is identical and proves (G.8).

Now set

$$(G.10) \quad x_{u,\tau}^\pm = q_\tau + \frac{\sqrt{n}}{k_u} \left\{ -\frac{Z_n(u, q_\tau)}{f(q_\tau)} \pm \varepsilon \right\}.$$

Since $k_u \geq u_0 n/2$, the two points in (G.10) lie within $O_{\mathbb{P}}(n^{-1/2})$ of q_τ . Taylor's theorem and the Lipschitz density condition give

$$(G.11) \quad \frac{k_u}{\sqrt{n}} \{F(x_{u,\tau}^+) - \tau\} = -Z_n(u, q_\tau) + \varepsilon f(q_\tau) + o_{\mathbb{P}}(1),$$

uniformly in u, τ ; the same calculation gives

$$\frac{k_u}{\sqrt{n}} \{F(x_{u,\tau}^-) - \tau\} = -Z_n(u, q_\tau) - \varepsilon f(q_\tau) + o_{\mathbb{P}}(1).$$

Lemmas 3.5 and 3.6 replace $Z_n(u, x_{u,\tau}^\pm)$ by $Z_n(u, q_\tau)$, while (G.6) replaces G_n by Z_n . Therefore, since $f(q_\tau) \geq c_f$, with probability tending to one,

$$(G.12) \quad F_{k_u}(x_{u,\tau}^-) \leq \tau - \frac{\varepsilon c_f}{2} < \tau \leq F_{k_u}(x_{u,\tau}^+)$$

uniformly over u, τ . With the convention $\hat{q}_{u,\tau} = \inf\{x : F_{k_u}(x) \geq \tau\}$, the right inequality in (G.12) gives $\hat{q}_{u,\tau} \leq x_{u,\tau}^+$. The strict left inequality gives $\hat{q}_{u,\tau} > x_{u,\tau}^-$; indeed, the empirical distribution function is right-continuous, so $F_{k_u}(\hat{q}_{u,\tau}) \geq \tau$, and $\hat{q}_{u,\tau} \leq x_{u,\tau}^-$ would imply $F_{k_u}(x_{u,\tau}^-) \geq \tau$ by monotonicity. Hence

$$x_{u,\tau}^- \leq \hat{q}_{u,\tau} \leq x_{u,\tau}^+.$$

Letting $\varepsilon \downarrow 0$ in (G.10) proves (3.10). $\square$

LEMMA G.1 (Sequential Bartlett covariance consistency). *Under the assumptions of Theorem 4.1,*

$$\max_{u,v \in \mathcal{U}_n} \max_{j,k \leq M_n} |\text{Cov}^*\{B_n^*(u, j), B_n^*(v, k)\} - (u \wedge v - uv)\Gamma(s_j, s_k)| = o_{\mathbb{P}}\{(\log(M_n K_n))^{-2}\}.$$

PROOF. Write $m = b_{\text{mul},n}$. Conditionally on the data,

$$(G.13) \quad \text{Cov}^*\{G_n^*(u, j), G_n^*(v, k)\} = \frac{1}{n} \sum_{i \leq [nu]} \sum_{\ell \leq [nv]} w_m(i - \ell) \hat{Y}_i(j) \hat{Y}_\ell(k),$$

where $w_m(h) = (1 - |h|/(m+1))\mathbf{1}(|h| \leq m)$. We first replace the centered variables $\hat{Y}_i$ by the population-centered variables Y_i . The algebraic identity is

$$(G.14) \quad \hat{Y}_i(j) \hat{Y}_\ell(k) - Y_i(j) Y_\ell(k) = -\bar{Y}_n(j) Y_\ell(k) - \bar{Y}_n(k) Y_i(j) + \bar{Y}_n(j) \bar{Y}_n(k).$$

For each fixed j , the sample mean $n^{-1} \sum_i Y_i(j)$ is a bounded geometrically dependent average. Applying the high-dimensional physical dependence maximal inequality of [Zhang and Wu \(2017\)](#) to the vector $\{Y_i(j) : j \leq M_n\}$ gives

$$(G.15) \quad \max_{j \leq M_n} |\bar{Y}_n(j)| = O_{\mathbb{P}}\{\sqrt{\log(M_n)/n}\}.$$

Since $\sum_{\ell} |w_m(i - \ell)| \leq Cm$ and $M_n K_n \leq n^C$, the centering cost obtained from (G.14) and (G.15) is

$$(G.16) \quad O_{\mathbb{P}}\{m\sqrt{\log(M_n)/n}\} = o_{\mathbb{P}}\{(\log(M_n K_n))^{-2}\}$$

for $m \asymp n^{1/4}$. Thus $\hat{Y}_i$ can be replaced by Y_i in (G.13) at the required rate.

It remains to control the population-centered covariance. Let $\mathcal{U}_n^+ = \mathcal{U}_n \cup \{1\}$. After writing $\ell = i + h$, the covariance expression with Y_i in place of $\hat{Y}_i$ is

$$\frac{1}{n} \sum_{|h| \leq m} w_m(h) \sum_{\substack{i \leq [nu] \\ i+h \leq [nv]}} Y_i(j) Y_{i+h}(k), \quad u, v \in \mathcal{U}_n^+,$$

up to the harmless convention that indices outside $\{1, \dots, n\}$ are discarded. Its expectation equals

$$(G.17) \quad \frac{|n(u \wedge v)|}{n} \sum_{|h| \leq m} w_m(h) \text{Cov}\{Y_0(j), Y_h(k)\} + O(m/n),$$

uniformly over $u, v \in \mathcal{U}_n^+$ and $j, k \leq M_n$. The difference between (G.17) and $(u \wedge v)\Gamma(s_j, s_k)$ is the sum of three terms: the integer-time edge $O(1/n)$, the boundary edge $O(m/n)$, and the Bartlett truncation bias. Geometric indicator dependence makes the truncation tail exponentially small in m , while

$$m/n = o\{(\log(M_n K_n))^{-2}\}.$$

For the stochastic fluctuation, define the product process

$$(G.18) \quad Z_i(h, j, k) = Y_i(j) Y_{i+h}(k) - \mathbb{E}\{Y_i(j) Y_{i+h}(k)\}.$$

The process in (G.18) inherits geometric physical dependence uniformly over $|h| \leq m$: replacing ε_0 changes the product by at most the sum of the two changed indicator coordinates. Applying [Zhang and Wu \(2017\)](#) to the bounded vector indexed by (h, j, k, u, v) , with $|h| \leq m$, $j, k \leq M_n$, and $u, v \in \mathcal{U}_n^+$, gives

$$(G.19) \quad \max_{u, v \in \mathcal{U}_n^+} \max_{j, k \leq M_n} \max_{|h| \leq m} \left| \frac{1}{n} \sum_{\substack{i \leq [nu] \\ i+h \leq [nv]}} Z_i(h, j, k) \right| = O_{\mathbb{P}}\left(\sqrt{\frac{\log(M_n K_n)}{n}}\right),$$

because the effective dimension satisfies

$$(G.20) \quad p_n \leq CmM_n^2 K_n^2 \leq n^C.$$

Summing the Bartlett weights over $|h| \leq m$ in (G.19) yields

$$(G.21) \quad O_{\mathbb{P}}\{m\sqrt{\log(M_n K_n)/n}\} = o_{\mathbb{P}}\{(\log(M_n K_n))^{-2}\}.$$

Combining (G.16), (G.17), and (G.21) gives the covariance consistency for $G_n^*(u, j)$ uniformly over $u \in \mathcal{U}_n^+$ and $j \leq M_n$. Applying the deterministic bridge linear transformation $G_n^*(u, j) - uG_n^*(1, j)$ gives the displayed covariance consistency for B_n^* . $\square$

PROOF OF THEOREM 4.1. Let $D_n = M_n K_n$ and list the coordinates $(u, j) \in \mathcal{U}_n \times \{1, \dots, M_n\}$. Conditionally on the data, $B_n^*(\mathcal{U}_n, \Theta_n)$ is a centered Gaussian vector in $\mathbb{R}^{D_n}$, and $B_n(\mathcal{U}_n, \Theta_n)$ is a centered Gaussian vector with the target bridge covariance. To handle the absolute maximum, pass to the doubled vectors

$$\tilde{B}_n^* = (B_n^*, -B_n^*), \quad \tilde{B}_n = (B_n, -B_n)$$

in $\mathbb{R}^{2D_n}$. Then $\max_{u,j} |B_n^*(u, j)| = \max_{\ell} \tilde{B}_n^*_{\ell}$, and similarly for B_n .

Lemma G.1 gives

$$\Delta_n = \max_{\ell, r \leq 2D_n} \left| \text{Cov}^*(\tilde{B}_{n,\ell}^*, \tilde{B}_{n,r}^*) - \text{Cov}(\tilde{B}_{n,\ell}, \tilde{B}_{n,r}) \right| = o_{\mathbb{P}}\{(\log D_n)^{-2}\}.$$

The target variances satisfy

$$\text{Var}\{B_n(u, j)\} = u(1-u)\Gamma(s_j, s_j) \geq u_0(1-u_0)c_{\Gamma}$$

for $u \in \mathcal{U}_n$, and the same lower bound holds for the conditional bootstrap variances with probability tending to one because of the covariance consistency above. The Gaussian comparison inequality of Chernozhukov, Chetverikov and Kato (2014), applied conditionally on the data to the two doubled Gaussian vectors, therefore gives

$$\sup_x \left| \mathbb{P}^*(\max_{\ell} \tilde{B}_{n,\ell}^* \leq x) - \mathbb{P}(\max_{\ell} \tilde{B}_{n,\ell} \leq x) \right| \leq C \Delta_n^{1/3} \{1 \vee \log(D_n/\Delta_n)\}^{2/3} = o_{\mathbb{P}}(1),$$

which is exactly (4.2). $\square$

APPENDIX H: PROOFS FOR MODEL-LEVEL VERIFICATION

PROOF OF PROPOSITION 5.1. The proof has two parts: a finite-net maximal bound derived from the physical dependence of the predictive state, and a discretization argument. Let $\varepsilon_n^{\partial} = n^{-1}$, and let $\mathcal{N}_n = \{w_1, \dots, w_{N_n}\}$ be an ε_n^{∂} -net of K_{A_n} in the sup norm. Since $A_n \leq n^C$,

$$(H.1) \quad N_n = |\mathcal{N}_n| \leq C(A_n/\varepsilon_n^{\partial})^d \leq n^C.$$

Fix j and define the N_n -dimensional vector

$$(H.2) \quad V_i^{(n,j)} = (\zeta_i^{(j)}(w_1), \dots, \zeta_i^{(j)}(w_{N_n})), \quad \zeta_i^{(j)}(w) = Q_j(w, H_{i-L}) - \mathbb{E}Q_j(w, H_0).$$

Let $s_n = 2 \vee \log N_n$. Then $|x|_{\infty} \leq |x|_{s_n} \leq e|x|_{\infty}$ for $x \in \mathbb{R}^{N_n}$, and the boundedness of Q_j gives $|V_i^{(n,j)}|_{\infty} \leq C$. With $\mathcal{F}_i = \sigma(\varepsilon_r : r \leq i)$ and $\mathcal{P}_i Z = \mathbb{E}(Z | \mathcal{F}_i) - \mathbb{E}(Z | \mathcal{F}_{i-1})$, the centered causal vector has the projection expansion

$$(H.3) \quad V_i^{(n,j)} = \sum_{k=0}^{\infty} \mathcal{P}_{i-k} V_i^{(n,j)}$$

in $L^v(\ell_{s_n}^{N_n})$. The convergence follows from the projection bound below and the summability of the state physical-dependence coefficients.

For a fixed k , the sequence $(\mathcal{P}_{i-k} V_i^{(n,j)})_{i=1}^n$ is an $\ell_{s_n}^{N_n}$ -valued martingale-difference sequence. The ℓ_s -valued martingale moment inequality used by Zhang and Wu (2017) gives

$$(H.4) \quad \left\| \sum_{i=1}^n \mathcal{P}_{i-k} V_i^{(n,j)} \right\|_{s_n} \Big\|_v \leq C_v \sqrt{ns_n} \left\| \mathcal{P}_0 V_k^{(n,j)} \right\|_{s_n} \Big\|_v.$$

Combining (H.3), (H.4), and Minkowski's inequality yields

$$(H.5) \quad \left\| \left\| n^{-1/2} \sum_{i=1}^n V_i^{(n,j)} \right\|_{\infty} \right\|_v \leq C_v \sqrt{s_n} \sum_{k=0}^{\infty} \left\| |\mathcal{P}_0 V_k^{(n,j)}|_{s_n} \right\|_v.$$

Let $V_k^{(n,j),*(0)}$ be obtained by replacing H_{k-L} by $H_{k-L}^{*(0)}$, with the convention that the difference is zero when $k < L$. Conditional Jensen and the standard projection-coupling comparison give

$$(H.6) \quad \left\| |\mathcal{P}_0 V_k^{(n,j)}|_{s_n} \right\|_v \leq C \left\| |V_k^{(n,j)} - V_k^{(n,j),*(0)}|_{s_n} \right\|_v.$$

The equivalence of ℓ_{s_n} and ℓ_{∞} , together with the state-Lipschitz assumption on Q_j , gives

$$(H.7) \quad \left\| |V_k^{(n,j)} - V_k^{(n,j),*(0)}|_{s_n} \right\|_v \leq C \left\| d_{\mathcal{H}}(H_{k-L}, H_{k-L}^{*(0)}) \right\|_v.$$

The summability assumption in Proposition 5.1 therefore implies

$$(H.8) \quad \max_{j \leq d} \left\| \max_{w \in \mathcal{N}_n} \left\| n^{-1/2} \sum_{i=1}^n \zeta_i^{(j)}(w) \right\| \right\|_v \leq C \sqrt{\log n}.$$

Thus the same finite-net maximum is $O_{\mathbb{P}}(\sqrt{\log n})$.

For any $w \in K_{A_n}$, choose $\pi w \in \mathcal{N}_n$ with $|w - \pi w|_{\infty} \leq \varepsilon_n^{\partial}$. By the Lipschitz condition on Q_j ,

$$(H.9) \quad \left| n^{-1/2} \sum_{i=1}^n \{ \zeta_i^{(j)}(w) - \zeta_i^{(j)}(\pi w) \} \right| \leq 2C \sqrt{n} \varepsilon_n^{\partial} = O(n^{-1/2}).$$

Taking the maximum over j and the supremum over $w \in K_{A_n}$ in (H.8)–(H.9) proves $R_n^{\partial}(A_n) = O_{\mathbb{P}}(\sqrt{\log n})$ for every deterministic $A_n \leq n^C$. If $b_n \downarrow 0$ and $\log n = O(nb_n)$, then the radius $A_n(b_n)$ in (2.8) is bounded by a fixed power of n . Hence the preceding bound verifies Condition 2.8 uniformly over all admissible sequences b_n , with $L_n^{\partial} = O(\sqrt{\log n})$. $\square$

PROOF OF PROPOSITION 5.2. Conditional on ξ_t , the only random component of X_{t+L} is $\eta_{t,L}$. Therefore

$$F_L(s | \xi_t) = \int_{(-\infty, s-H_t]} p_{\eta}(z) dz.$$

Differentiating along coordinate j gives

$$(H.10) \quad q_{L,j}(w | \xi_t) = Q_j(w, H_t) = \int_{(-\infty, w_{-j}-H_{t,-j}]} p_{\eta}(w_j - H_{t,j}, z_{-j}) dz_{-j}.$$

The bounded coordinate marginal densities of $\eta_{t,L}$ imply Condition 2.6. The assumed boundedness and Lipschitz property of Q_j in (H.10), together with the geometric or summable polynomial physical dependence of H_t , verify Proposition 5.1; hence Condition 2.8 holds with an admissible L_n^{∂} of order $\sqrt{\log n}$.

For the physical-dependence coefficient, use the same future innovation $\eta_{k,L}$ in the original and coupled versions. Then

$$|X_{k+L} - X_{k+L}^{*(0)}|_{\infty} \leq |H_k - H_k^{*(0)}|_{\infty}.$$

The displayed geometric or polynomial state-dependence bound therefore transfers directly to X_t . The moment and marginal regularity assumptions complete the verification of the corresponding main theorem. $\square$

PROOF OF PROPOSITION 5.3. Let $X_k^{*(0)}$ be the coupled copy obtained by replacing the time-zero innovation in the causal input. Use the same current innovation ε_k in both copies. If G is Lipschitz in the state with constant L_G , then

$$(H.11) \quad |X_k - X_k^{*(0)}|_\infty \leq L_G d_{\mathcal{H}}(H_k, H_k^{*(0)}).$$

Taking L^q -norms in (H.11) gives

$$(H.12) \quad \delta_q(k) \leq L_G \left\| d_{\mathcal{H}}(H_k, H_k^{*(0)}) \right\|_q.$$

Thus the geometric or polynomial state-dependence rate transfers directly to X_k . The assumed predictive boundary densities give Conditions 2.6, 2.7, and, by Proposition 5.1, Condition 2.8.

For the linear state $H_t = \sum_{\ell \geq 0} A_\ell \zeta_{t-\ell}$, the replacement of ζ_0 affects H_k only through the coefficient A_k . Hence

$$(H.13) \quad \left\| |H_k - H_k^{*(0)}|_\infty \right\|_q \leq \|A_k\|_{\infty \rightarrow \infty} \left\| |\zeta_0 - \zeta'_0|_\infty \right\|_q.$$

The coefficient tail in (H.13) gives the corresponding physical-dependence rate. The summability condition

$$\sum_{\ell \geq 0} \|A_\ell\|_{\infty \rightarrow \infty} < \infty$$

ensures that the state is well defined in the sup norm. $\square$

PROOF OF PROPOSITION 5.4. Construct the coupled state by replacing the time-zero innovation and then using the same future innovations in the two recursions. By the contraction assumption,

$$(H.14) \quad d_{\mathcal{H}}(H_k, H_k^{*(0)}) \leq \rho^k d_{\mathcal{H}}(H_0, H_0^{*(0)}).$$

The finite q th moment of the stationary state gives

$$(H.15) \quad \left\| d_{\mathcal{H}}(H_k, H_k^{*(0)}) \right\|_q \leq C \rho^k.$$

Since $X_t = G(H_t, \varepsilon_t)$ is Lipschitz in H_t , using the same innovation ε_k in the two copies gives

$$(H.16) \quad |X_k - X_k^{*(0)}|_\infty \leq L_G d_{\mathcal{H}}(H_k, H_k^{*(0)}).$$

Taking L^q -norms in (H.16) and using (H.15) yields

$$(H.17) \quad \delta_q(k) = \left\| |X_k - X_k^{*(0)}|_\infty \right\|_q \leq C \rho^k.$$

The finite-step predictive distribution assumptions give the slab and boundary differentiability conditions, and Proposition 5.1 gives local boundary predictability. $\square$

PROOF OF PROPOSITION 5.6. Use the same innovation ε_k in X_k and $X_k^{*(0)}$. Then

$$(H.18) \quad |X_k - X_k^{*(0)}|_\infty \leq |\varepsilon_k|_\infty |\sigma_k - \sigma_k^{*(0)}|_\infty.$$

Because ε_k is independent of the past volatility state, Holder's inequality gives

$$(H.19) \quad \delta_q(k) \leq \left\| |\varepsilon_0|_\infty \right\|_{2q} \left\| |\sigma_k - \sigma_k^{*(0)}|_\infty \right\|_{2q}.$$

Equation (H.19) transfers the assumed geometric or polynomial volatility dependence to the physical-dependence coefficient of X_k .

Conditional on σ_t , the coordinates of X_t are independent and the j th coordinate has density

$$(H.20) \quad x \mapsto \sigma_{t,j}^{-1} f_{\varepsilon_j}(x/\sigma_{t,j}).$$

The lower bound $\sigma_{t,j} \geq c_\sigma$ and the bounded innovation density in (H.20) imply the finite-step slab anti-concentration condition. The coordinate boundary derivative is

$$(H.21) \quad q_{1,j}(z, u_{-j} \mid \sigma_t) = \sigma_{t,j}^{-1} f_{\varepsilon_j}(z/\sigma_{t,j}) \prod_{\ell \neq j} F_{\varepsilon_\ell}(u_\ell/\sigma_{t,\ell}).$$

Bounded first derivatives of the innovation densities, the bounds $\sup_x |x f'_{\varepsilon_j}(x)| < \infty$, and the lower volatility bound make (H.21) Lipschitz in (z, u_{-j}) and in the volatility state on each localization cube. Proposition 5.1 therefore verifies the boundary predictability condition. $\square$